Lecture · State Space Models

Kalman Filter and EM

State space models via spending, rates, ETF hedging, volatility and nowcasting

State Space Model · Kalman Filter · Expectation-Maximization

Two equations describe a hidden state, the Kalman filter estimates the state, and EM learns the model's numbers.

Two equations and two tools sum up the model

Three messages that run through the whole lecture

1

Two equations define the model

The state equation says how a hidden quantity moves over time; the observation equation says how it shows up in data.

2

The Kalman filter estimates states; EM estimates parameters

The filter updates the state with each observation; EM alternates between estimating the state (E) and fixing Q and R (M).

3

The same framework covers finance and economics

Spending, interest rates, hedge ratios, volatility and the business cycle are all treated as hidden states and estimated the same way.

01

PART 1 · Basic model

Two basic equations

State equation, observation equation, Kalman filter, EM

A state space model consists of two equations

We start with the case where the state and the observation are single numbers

State equation

$$\underbrace{x_t}_{\text{today's state}} = \underbrace{A}_{\text{carry-over}} \underbrace{x_{t-1}}_{\text{yesterday's state}} + \underbrace{w_t}_{\text{state change}}, \qquad w_t \sim N(0, Q)$$

Observation equation

$$\underbrace{y_t}_{\text{observation}} = \underbrace{H_t}_{\text{link number}} \underbrace{x_t}_{\text{state}} + \underbrace{v_t}_{\text{obs. residual}}, \qquad v_t \sim N(0, R)$$

What the state equation says

- How the quantity we want to know changes over time

What the observation equation says

- How that quantity relates to the data we actually see

Each symbol has a fixed role

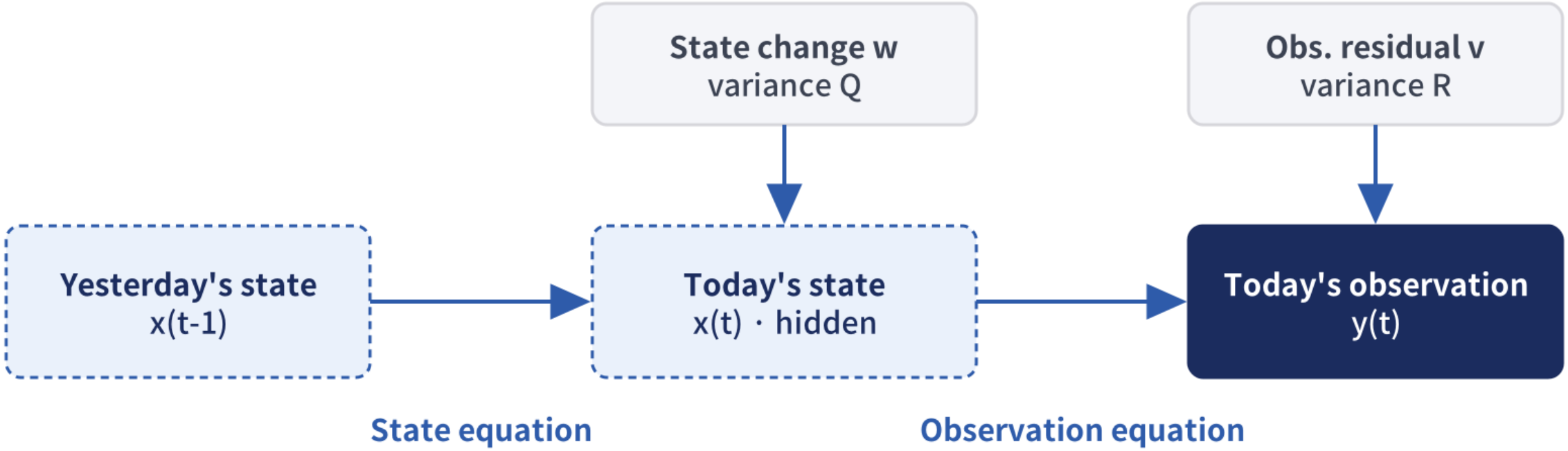
Variance is the size of fluctuation · both errors have mean 0 and are independent

Symbol	Meaning
x_t	The state we want to know; hard to observe exactly
y_t	The data we actually have
A	How yesterday's state carries over to today
H_t	Number linking the state to the observation (may vary with known inputs)
w_t	New change in the state itself
v_t	Part of the observation the state does not explain
Q	Variance of the state change
R	Variance of the observation residual

The mean and variance of the initial state must also be set

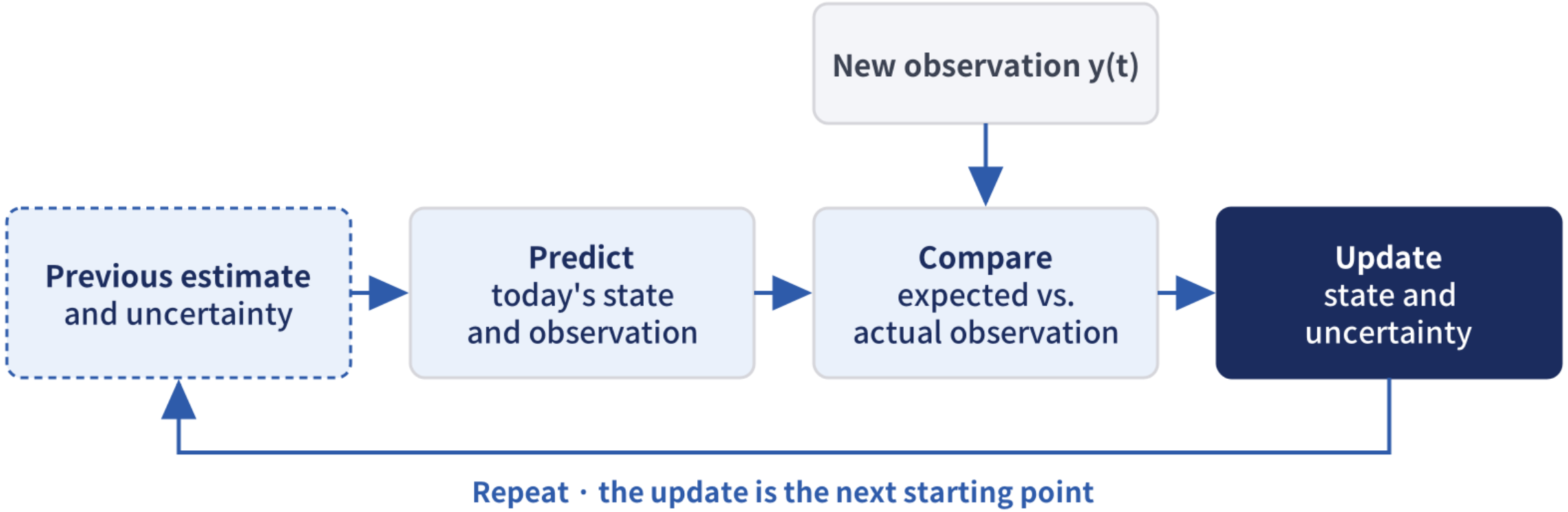
The hidden state persists and generates observations

The state may be an unknown value like true spending, or a coefficient like a hedge ratio



The Kalman filter updates the state each step

Even with A, H, Q, R and the initial mean and variance known, the actual state must still be estimated



Seven lines of predict, compare and update

Minus superscript: forecast before observing · hat: estimate after · P: error variance

$$\hat{x}_t^- = A\hat{x}_{t-1} \qquad \text{state forecast}$$

$$P_t^- = A^2 P_{t-1} + Q \qquad \text{forecast error variance}$$

$$e_t = y_t - H_t\hat{x}_t^- \qquad \text{innovation (actual - expected)}$$

$$S_t = H_t^2 P_t^- + R \qquad \text{innovation variance}$$

$$K_t = \frac{P_t^- H_t}{S_t} \qquad \text{Kalman gain}$$

$$\hat{x}_t = \hat{x}_t^- + K_t e_t \qquad \text{state update}$$

$$P_t = (1 - K_t H_t) P_t^- \qquad \text{updated error variance}$$

Inputs and outputs

- Input: previous estimate and its uncertainty, today's observation
- Output: today's estimate with smaller uncertainty
- Innovation e is actual minus expected; gain K decides how much of it to use
- With H = 1, K is a weight between forecast and observation

EM alternates estimating states and fixing numbers

We fix A and the observation structure and mainly learn Q and R · E is expectation, M is maximization

What we want	Method
The current state	Kalman filter
Past states re-estimated with the full record	Kalman smoother
Variances Q and R of state change and observation residual	EM

E-step · estimate states with current numbers

- Run the filter forward and the smoother backward with the current Q and R
- Get the mean and uncertainty of the state at each time

M-step · fix numbers to fit the states

- Take the average size of the residuals as the new variance
- The new Q and R come out here

Q and R are E-step inputs and M-step outputs

E = Expectation, M = Maximization · each round produces new Q and R

Step	Input	What it does	Output
E-step	Current Q and R, full observation record	Runs the filter forward and the smoother backward	State mean and uncertainty at each time, expected squared residuals
M-step	Expected squared residuals from the E-step	Averages them over time	New Q and new R

After one round

- Go back to the E-step with the new Q and R; stop when the likelihood gain is small enough

Expected squared residuals link the E-step and M-step

Since the state is not known exactly, residuals are computed as average sizes, not single values

What the E-step produces

Considering all plausible states, it computes the average squared residual

Two kinds of residuals

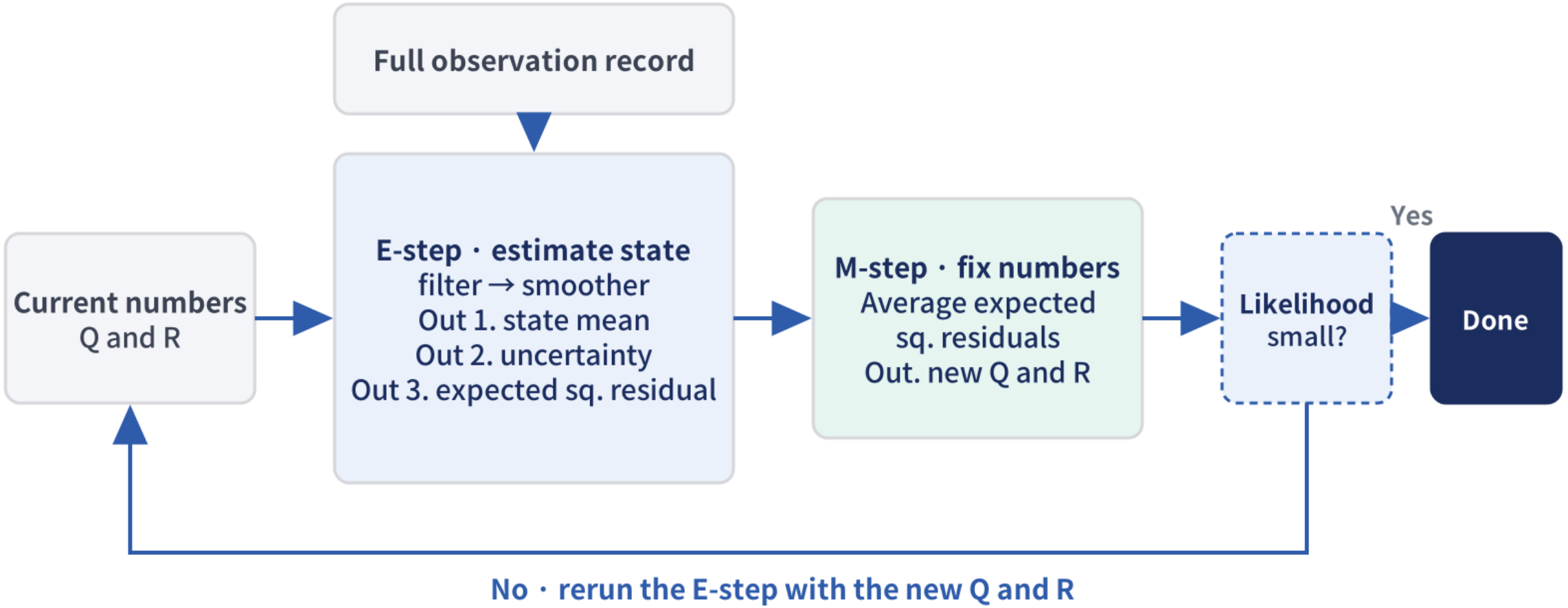
Observation minus state is used for R; today's minus yesterday's state is used for Q

What the M-step does

Averaging the two over time gives the new R and the new Q directly

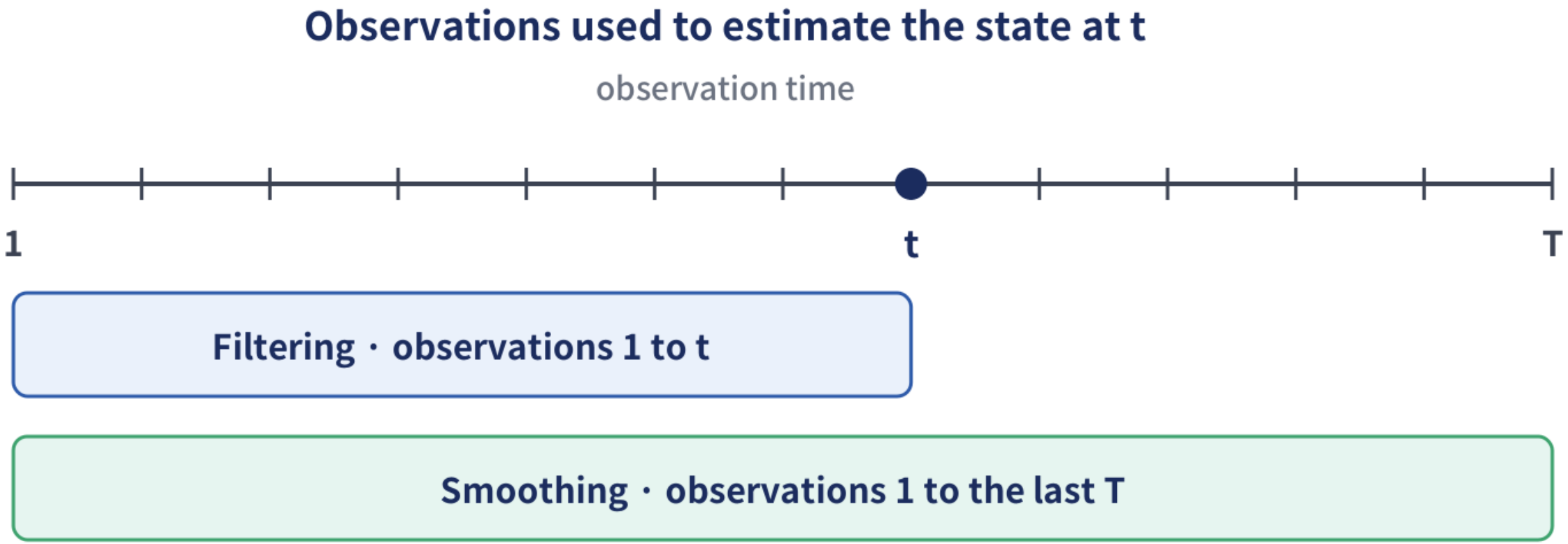
Each new set of numbers triggers new state estimates

EM may converge differently from other starting values · check results and likelihood



Filtering and smoothing use different observations

EM does not treat the state estimate as the truth; it accounts for all plausible values



Fix the big picture and Claude Code does the math

Labs run in the Claude Code desktop app (Code tab) · students paste a prompt and check the results

1. Fix the big picture

Use the two equations to decide what is the hidden state, what is observed and which numbers to learn

2. Hand it over in a prompt

Claude Code downloads data, implements filter, smoother and EM, draws charts and fixes errors

3. Check the results

Did Q and R converge, does the estimate make sense, is future information kept out?

02

PART 2 · Spending example

Estimating true average spending

The simplest state space model with A = H = 1 · unit 10,000 KRW

True spending is the state; the survey is observed

A = H = 1 · unit: 10,000 KRW

State equation

$$\underbrace{x_t}_{\text{true avg. spending}} = x_{t-1} + \underbrace{w_t}_{\text{spending change}}, \qquad w_t \sim N(0, Q)$$

Observation equation

$$\underbrace{y_t}_{\text{survey average}} = x_t + \underbrace{v_t}_{\text{survey error}}, \qquad v_t \sim N(0, R)$$

State · true monthly average spending of all households

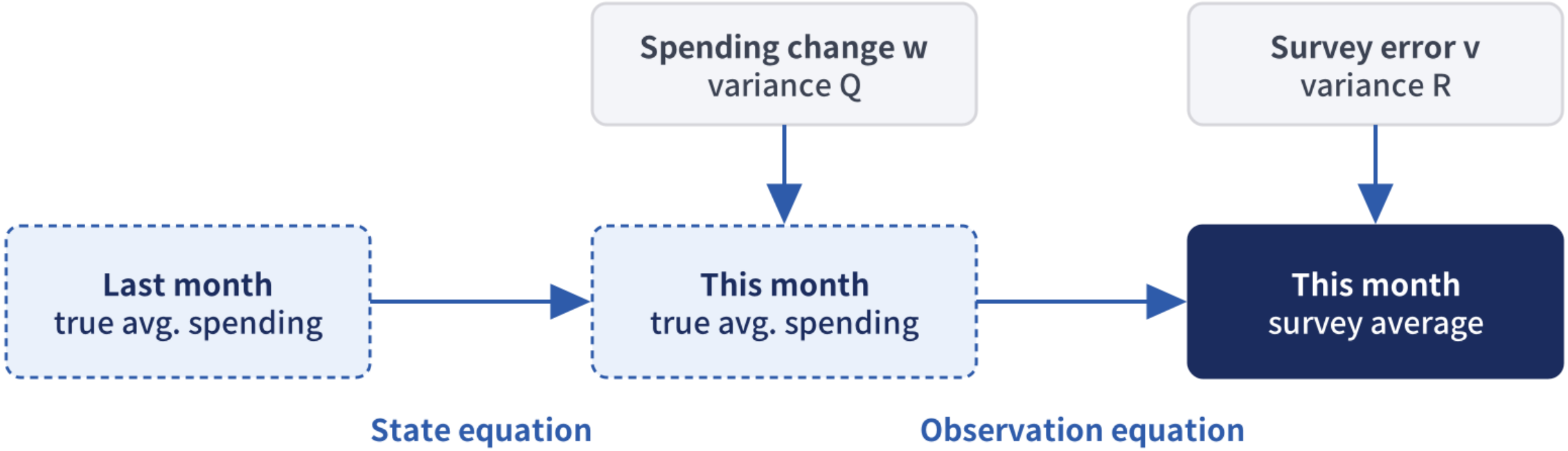
- Changes with income and spending needs

Observation · average from a household survey

- Can differ from the true average depending on the sampled households

Survey error makes the survey miss the true average

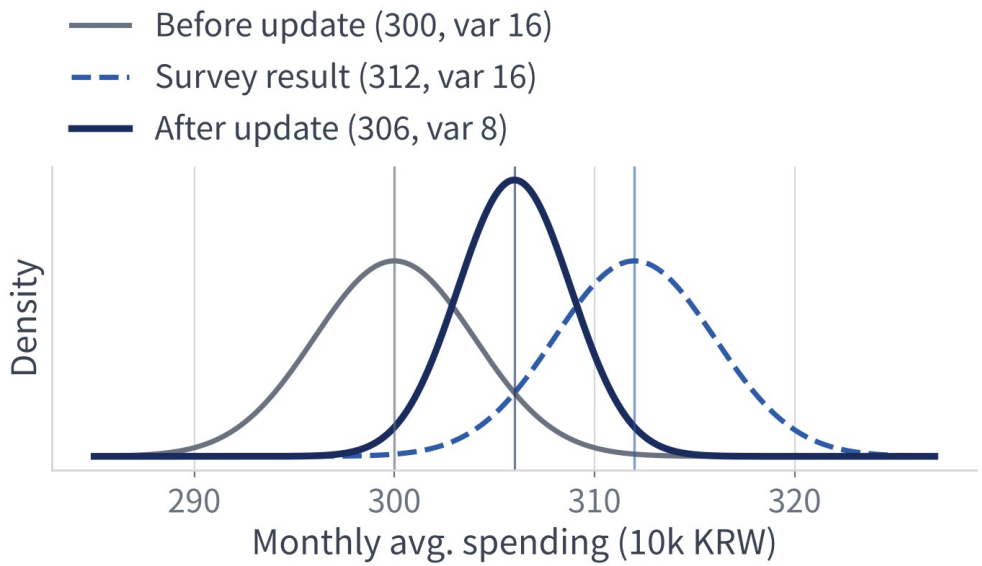
Q is the variance of true spending changes, R the variance of survey error



A survey of 312 lifts the estimate to 306

Hypothetical example · variance unit (10,000 KRW)^2 · step-by-step calculation in the workbook

Item	Value
Last month's updated estimate	300
Last month's updated error variance	8
Variance of true change Q	8
This month's survey result	312
Survey error variance R	16

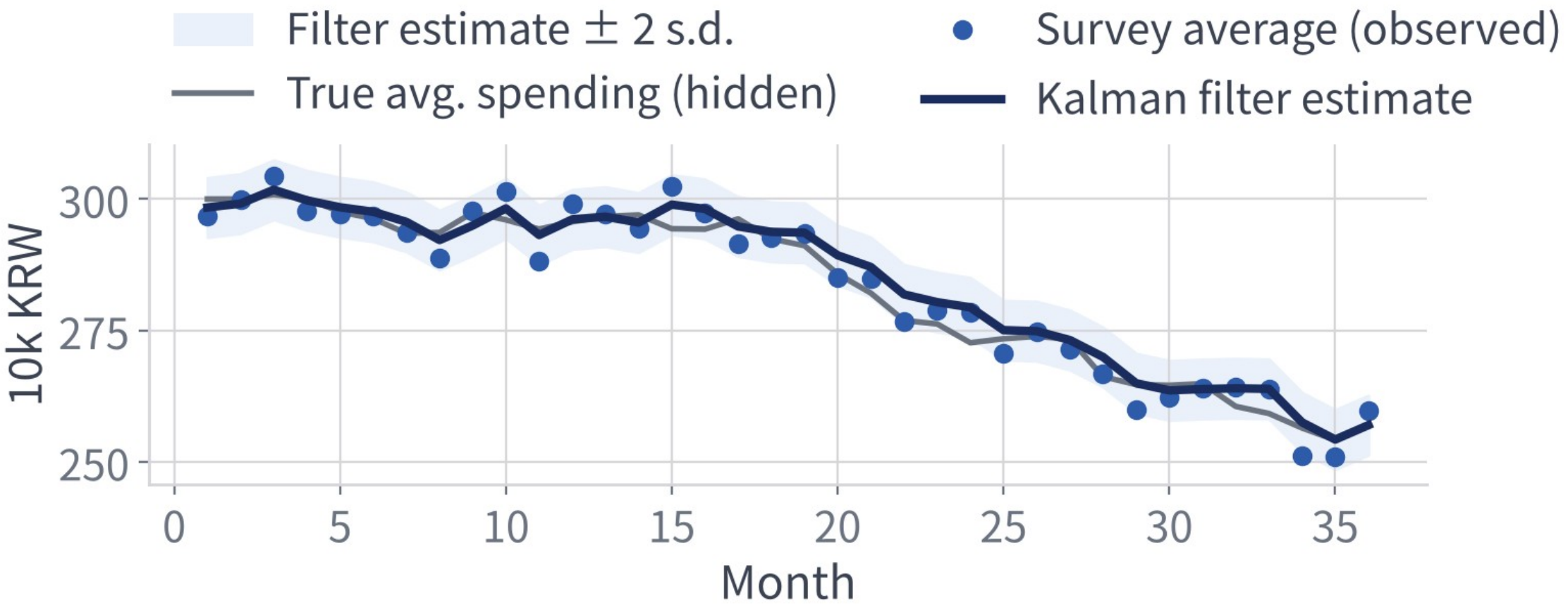


Result · error variance falls from 16 before to 8 after the observation

$$\hat{x}_t = 300 + 0.5 \times 12 = 306, \qquad P_t : \ 16 \ \rightarrow \ 8$$

The filter tracks true spending through noisy surveys

36-month simulated path with Q = 8, R = 16 · each month restarts from the update · ※ illustrative data



The E-step passes on each month as a range

m is the smoothed mean, V the uncertainty (smoothed variance) · ※ three illustrative months

Month	Survey	Smoothed mean	Uncertainty	Squared gap + uncertainty
1	312	308	6	16 + 6 = 22
2	305	304	5	1 + 5 = 6
3	298	300	6	4 + 6 = 10

E-step output · expected squared residual used for R

$$E[(y_t - x_t)^2 \mid Y] = \underbrace{(y_t - m_t)^2}_{\text{gap from smoothed mean}} + \underbrace{V_t}_{\text{remaining uncertainty}}$$

Because spending is a range, not a point, the residual grows by that uncertainty

V appears naturally when the expectation is computed

The E-step averages squared residuals without knowing true spending

A fact that holds for any quantity

$$\text{mean of squares} = (\text{mean})^2 + \text{variance}$$

Applied to the residual

$$E[(y_t - x_t)^2 \mid Y] = \underbrace{(y_t - m_t)^2}_{\text{gap from smoothed mean}} + \underbrace{V_t}_{\text{remaining uncertainty}}$$

Squared mean

- Squared gap between survey and smoothed mean

Variance

- Uncertainty V of true spending

Check with 1 and 3

- Mean of squares 5 = squared mean 4 + variance 1

The M-step averages expected residuals into new Q, R

All expectations were computed by the E-step with the current Q and R · T−1 monthly changes

M-step output · new Q

$$\underbrace{Q_{\text{new}}}_{\text{state change var.}} = \frac{1}{T-1}\sum_{t=2}^{T} E[(x_t - x_{t-1})^2 \mid Y]$$

M-step output · new R

$$\underbrace{R_{\text{new}}}_{\text{survey error var.}} = \frac{1}{T}\sum_{t=1}^{T} E[(y_t - x_t)^2 \mid Y]$$

Value used for Q

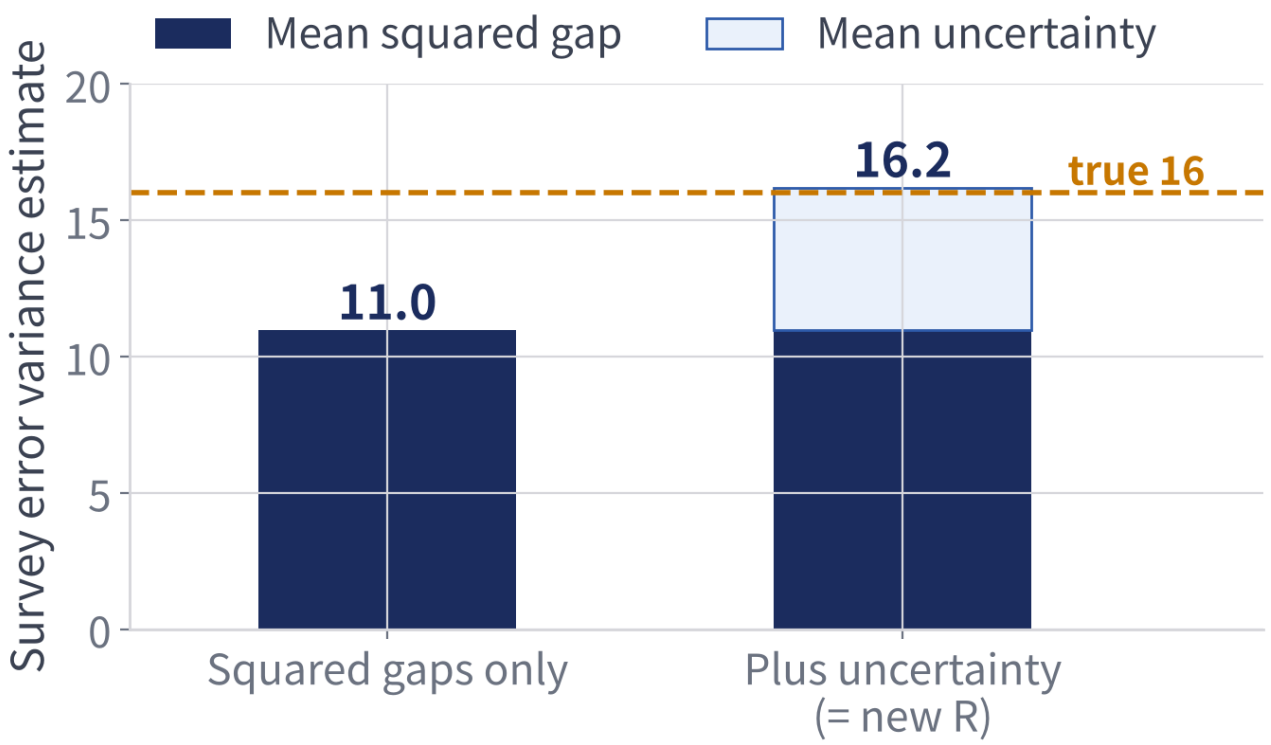
- Expected square of the month-to-month change
- Gap between two smoothed means plus both months' uncertainty and how the two estimates move together

Value used for R

- Expected square of survey minus true spending
- Squared gap from the smoothed mean plus that month's uncertainty

Dropping V underestimates R by the average V

240 months simulated with Q = 8, R = 16 · after EM convergence · ※ illustrative data

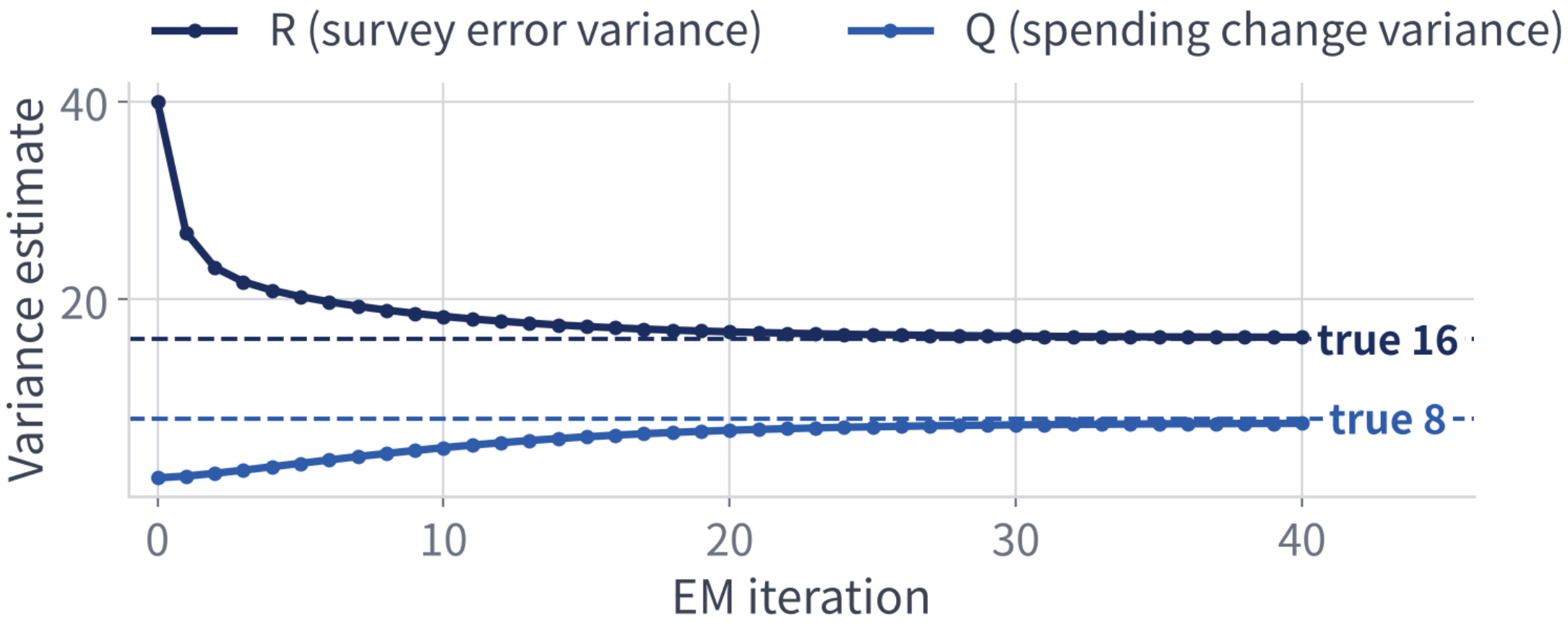


Why

- V is not a correction; it arises naturally from the expectation
- Dropping it treats true spending as a known point
- V is always positive, so R shrinks by that much
- A smaller R makes the next E-step trust the survey too much

As E and M repeat, Q and R settle near the true values

Starting from Q = 2, R = 40 · same 240-month illustrative data · ※ illustrative data



Lab · Solve the spending example with Claude Code

Big picture: true spending is the state, the survey is observed, Q and R are learned

[Input] Prompt · Claude Code desktop app · Code tab

```
Run a state space model lab. Work folder: ssm_lab/consumption.

1. Model: state x(t) = x(t-1) + w, observation y(t) = x(t) + v
   (state = true average spending, observation = survey average)
2. Simulate 120 months with Q=8, R=16, starting at 300.
3. Implement the Kalman filter, smoother and EM in numpy.
   Start EM at Q=2, R=40 and learn Q and R. For each iteration,
   log the E-step outputs and the M-step outputs (new Q, R).
4. Three charts: true / survey / filtered / smoothed paths;
   filter variance P vs smoothed variance V; Q and R by iteration.
5. Summarize in report.md and open it with the charts.
   If anything fails, fix it yourself and run to the end.
```

Output and checks

- Output: simulated data, filter/smoother/EM code, three charts, report.md
- Do Q and R converge near the true 8 and 16?
- Is V below P, and equal to it in the last month?
- Does the log separate E-step outputs from M-step outputs?

03

PART 3 · Interest rate example

Estimating a latent common rate level

A statistical model splitting rates into a persistent level and temporary deviations

Split market rates into level and temporary deviation

The observation is the daily market rate of a fixed maturity

State equation

$$\underbrace{\ell_t}_{\text{rate level}} = \ell_{t-1} + \underbrace{w_t}_{\text{level change}}, \qquad w_t \sim N(0, Q)$$

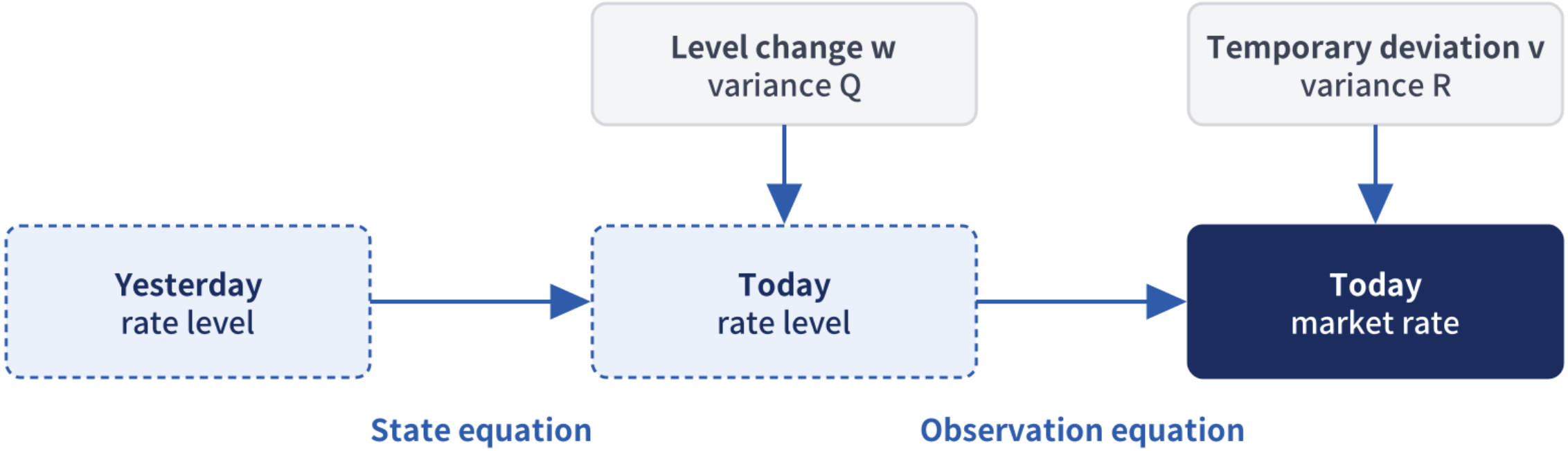
Observation eq.

$$\underbrace{y_t}_{\text{market rate}} = \ell_t + \underbrace{v_t}_{\text{temporary deviation}}, \qquad v_t \sim N(0, R)$$

Part	Meaning
ℓ_t	Rate level that persists over many days
w_t	New change that moves the level
v_t	Part of the observed rate that deviates from the level

Same structure as spending, with a latent level

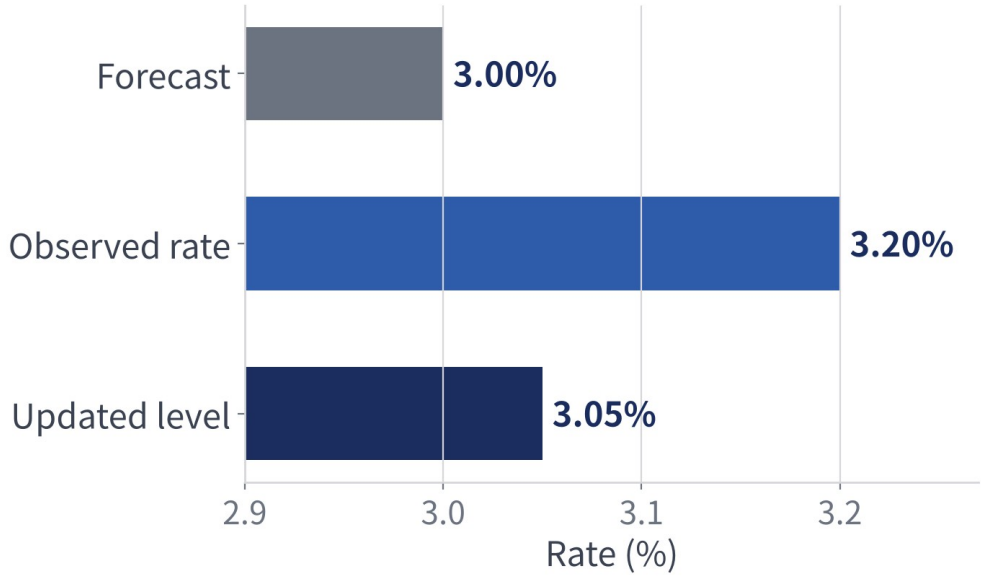
Spending had a real target, the true average; here the model defines the persistent part as the level



A 0.20%p rise moves the level estimate up 0.05%p

Hypothetical example · 3.00 means 3.00% · variance unit $(\%p)^2$

Item	Hypothetical value
Forecast of today's level	3.00%
Forecast error variance	0.0025
Observation residual variance R	0.0075
Observed rate	3.20%



Result

$$K_t = \frac{0.0025}{0.0025 + 0.0075} = 0.25, \qquad \hat{\ell}_t = 3.00 + 0.25(3.20 - 3.00) = 3.05\%$$

In the rate EM, E-step outputs are M-step inputs

Putting the rate level where true spending was gives the same calculation

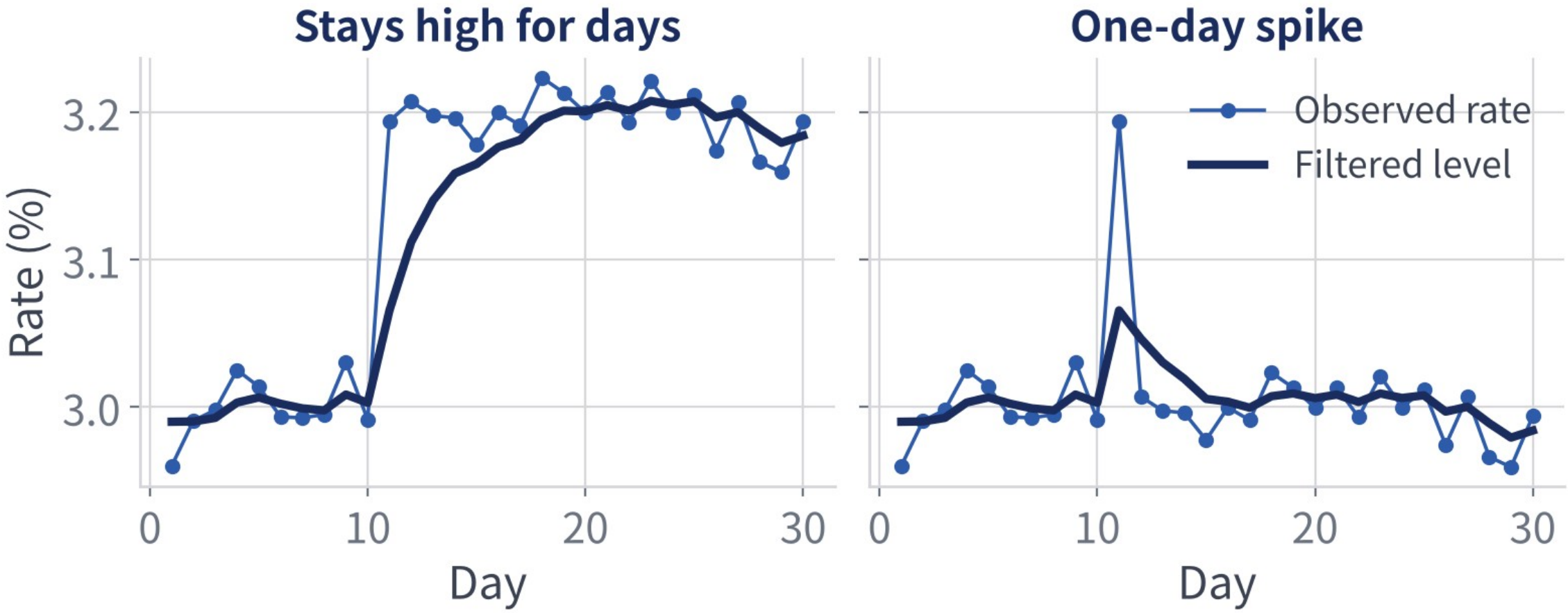
Step	Input	Output
E-step	Current Q and R, rate history	Mean and uncertainty of the level each day, two kinds of expected squared residuals
M-step	Expected squared residuals	New Q: average size of daily level changes
M-step	Expected squared residuals	New R: average size of deviations from the level

Intuition

- Many lasting changes raise Q; many one-day spikes that revert raise R

Whether a rise persists separates Q from R

Lasting moves signal level change; reverting spikes signal deviation · ※ illustrative data



Lab · Estimate the level of real interest rates

Big picture: the latent rate level is the state, the market rate is the observation, Q and R are learned

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```
Run the interest rate lab. Work folder: ssm_lab/rate.

1. Download the daily US 10-year Treasury yield (DGS10)
   from FRED since 2015 and drop missing days.
2. Model: level l(t) = l(t-1) + w, observation y(t) = l(t) + v
3. Implement the Kalman filter, smoother and EM in numpy to
   learn Q and R, and compare with statsmodels
   UnobservedComponents (local level).
4. Charts: observed rate with filtered and smoothed level;
   a zoom around days with large moves.
5. Interpret the Q/R ratio in one paragraph in report.md
   and open it with the charts. Fix any errors yourself.
```

Output and checks

- Output: rate data, EM vs statsmodels table, two charts, report.md
- Are the EM and statsmodels estimates of Q and R close?
- Does the level move a lot for lasting rises and little for one-day spikes?

04

PART 4 · EWC–EWA hedging

Updating the hedge ratio daily

Teaching example: hold EWC (Canada equity ETF) and short EWA (Australia equity ETF)

The hedge ratio is the state; EWC return is observed

Returns are used and the intercept is fixed at 0 · USD returns over the same trading interval

State equation

$$\underbrace{\beta_t}_{\text{hedge ratio}} = \beta_{t-1} + \underbrace{w_t}_{\text{ratio change}}, \qquad w_t \sim N(0, Q)$$

Observation equation

$$\underbrace{r_{\text{EWC},t}}_{\text{EWC return}} = \beta_t \underbrace{r_{\text{EWA},t}}_{\text{EWA return}} + \underbrace{v_t}_{\text{EWC own move}}, \qquad v_t \sim N(0, R)$$

Purpose

- Offset the part of EWC returns that moves with EWA by shorting EWA

Nature of the state

- Prices and returns are observed, but the ratio linking them must be estimated

The basic symbols map onto the hedging problem

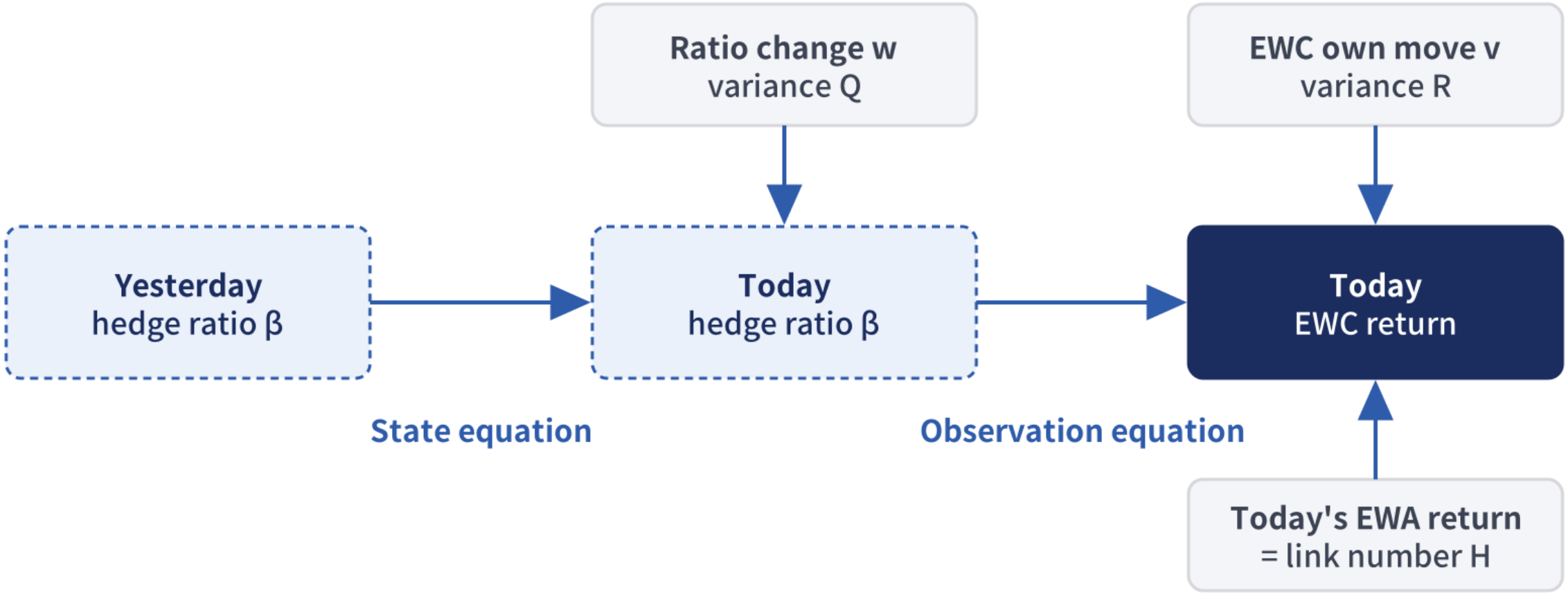
The day's observed EWA return takes the place of the link number in the observation equation

Basic model symbol	Meaning in this example
x_t	Hedge ratio
y_t	EWC return
H_t	Observed EWA return
Q	How much the hedge ratio changes over time
R	Variance of EWC return not explained by EWA

The state is a coefficient describing the return relationship

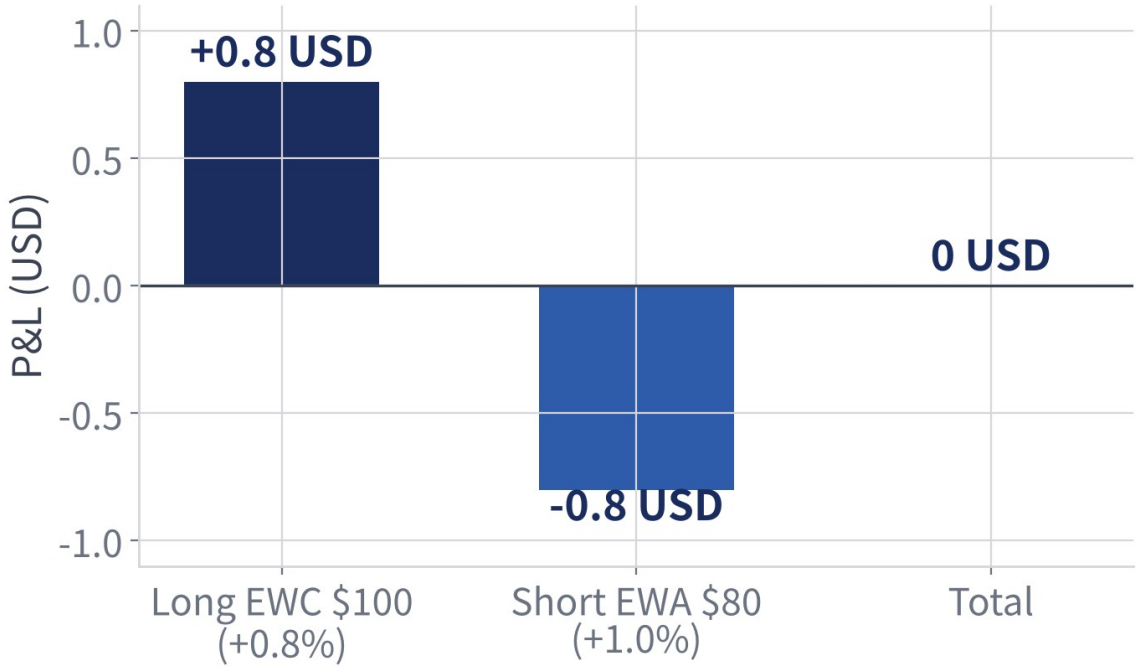
The EWA return acts as the link number

Today's hedge ratio and today's EWA return together explain today's EWC return



A hedge ratio of 0.8 means shorting $80 per $100

In a return regression the hedge ratio is a dollar ratio · hypothetical returns



Converting to share counts

$$\text{EWA shares to short} = \beta \times \text{EWC shares held} \times \frac{\text{EWC price}}{\text{EWA price}}$$

What remains in actual P&L

- Observation residual, trading costs, shorting costs
- This relation alone does not guarantee mean reversion or profit

A larger-than-expected EWC rise lifts the ratio to 1.00

Hypothetical example · returns in decimals (1% = 0.01) · expected +0.8%, actual +1.2%

Item	Value
Hedge ratio forecast	0.80
Forecast error variance	0.04
EWA return (link number)	0.010
Actual EWC return	0.012
Observation residual variance R	0.000004

The gain is not a weight here

- When the link number is not 1, K need not lie between 0 and 1
- Judge the update size by K times the innovation

Result · updated error variance falls from 0.04 to 0.02

$$K_t = \frac{0.04 \times 0.010}{0.010^2 \times 0.04 + 0.000004} = 50, \qquad \hat{\beta}_t = 0.80 + 50 \times 0.004 = 1.00$$

Update the ratio after observing, apply it next

On days with a negative EWA return, compute the gap and the direction of the update together

1. Today's interval

Use the ratio set with information up to yesterday

2. Interval ends

Observe the returns of both ETFs

3. Filter update

The Kalman filter updates the ratio to 1.00

4. Next tradable time

Adjust to shorting $100 of EWA per $100 of EWC

In hedging EM, the E-step passes on ratio ranges

EWA returns are treated as observed inputs · expectations use the current Q and R

Step	Output
E-step	Daily hedge ratio mean and uncertainty, link between neighboring days, two kinds of expected squared residuals
M-step · new Q	Average of the expected squared daily change of the hedge ratio
M-step · new R	Average of the expected squared observation residual

$$\underbrace{Q_{\text{new}}}_{\text{ratio change var.}} = \frac{1}{T-1}\sum_{t=2}^{T} E[(\beta_t - \beta_{t-1})^2 \mid \text{all data}]$$

$$\underbrace{R_{\text{new}}}_{\text{obs. residual var.}} = \frac{1}{T}\sum_{t=1}^{T} E[(r_{\text{EWC},t} - \beta_t r_{\text{EWA},t})^2 \mid \text{all data}]$$

The E-step's R value includes hedge ratio uncertainty

Mean of squares = squared mean + variance · so ratio uncertainty enters via V

$$\underbrace{(r_{\mathrm{EWC},t} - m_t r_{\mathrm{EWA},t})^2}_{\text{residual at smoothed mean}} + \underbrace{r_{\mathrm{EWA},t}^2 V_t}_{\text{hedge ratio uncertainty}}$$

Large Q

- Allows more room for the ratio to actually change

Large R

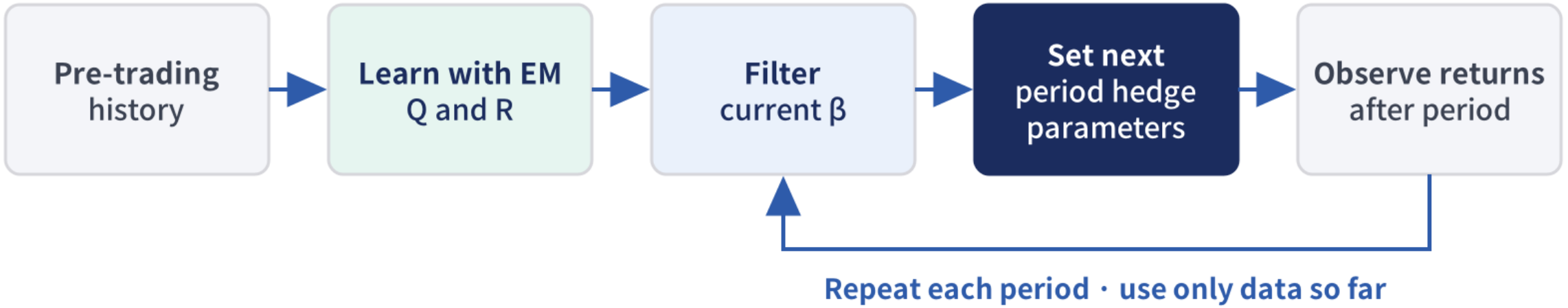
- Treats more of the daily gap between the ETFs as idiosyncratic

EWA return near 0

- That day says little about the ratio; at exactly 0 the gain is 0

At each trade date, use only data available then

Use the smoother only for EM on past data · never evaluate trades with smoothed ratios



Lab · Backtest the dynamic EWC–EWA hedge

Big picture: hedge ratio is the state, EWC return observed, EWA return the link

[Input] Prompt · Claude Code desktop app · Code tab

```
Run the EWC–EWA dynamic hedging lab. Work folder: ssm_lab/hedge.

1. Download EWC and EWA adjusted prices since 2012 with
   yfinance and compute daily returns.
2. Model: ratio b(t) = b(t-1) + w
          r_EWC(t) = b(t) * r_EWA(t) + v   (no intercept)
3. Learn Q and R with EM on 2012-2018, then run only the
   filter from 2019 on to update the ratio every day.
4. Backtest: today's position uses only the ratio known
   yesterday (no smoothed ratios). Compare hedged-return
   volatility with a fixed OLS ratio.
5. Charts: dynamic vs fixed ratio; cumulative P&L of both.
   Summarize in report.md and open it. Fix errors yourself.
```

Output and checks

- Output: training Q and R, daily hedge ratios, backtest table, two charts
- Is the position free of future information?
- Did the dynamic hedge lower hedged volatility? If not, is Q too large or too small?

05

PART 5 · Stochastic volatility

Estimating market turbulence as a state

Stochastic Volatility (SV) model · quasi-maximum likelihood approximation

The SV model treats log variance as the hidden state

Daily returns with the mean removed · the two shocks are independent of each other and over time

State equation

$$\underbrace{h_t}_{\text{log variance}} = \underbrace{\mu_h}_{\text{long-run mean}} + \underbrace{\phi}_{\text{persistence}}(h_{t-1} - \mu_h) + \underbrace{\eta_t}_{\text{vol shock}}, \qquad \eta_t \sim N(0, Q)$$

Observation equation

$$\underbrace{r_t}_{\text{return}} = \underbrace{\exp(h_t/2)}_{\text{volatility}} \underbrace{\varepsilon_t}_{\text{std. shock}}, \qquad \varepsilon_t \sim N(0, 1)$$

What we want to know

- How much the market can move today

The SV view

- Even one stock has calm and turbulent periods, so volatility changes daily

SV symbols cover returns, volatility and two shocks

Persistence below 1 means reversion to the long-run mean; near 1, shocks last longer

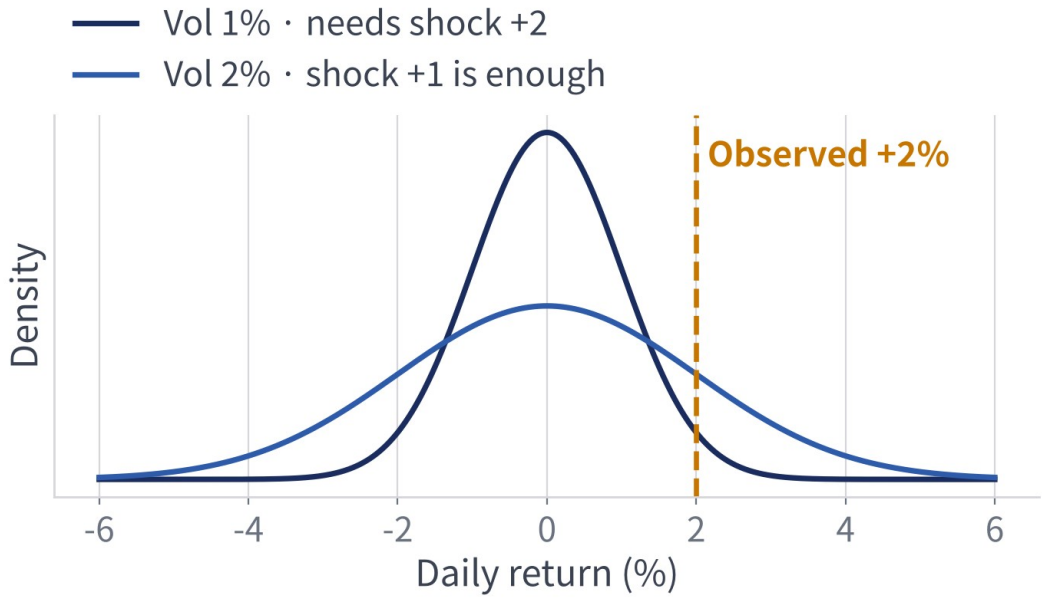
Symbol	Meaning
r_t	Observed return (1% = 0.01)
h_t	Hidden state: log variance
$\sigma_t = \exp(h_t/2)$	Daily volatility · conditional s.d. of returns
μ_h	Long-run mean of log variance
ϕ	How long high or low log variance persists
η_t	New shock to the volatility state
Q	Variance of the log variance shock
ε_t	Standardized shock setting the day's return sign and size

One day's return cannot pin down volatility

Two hypothetical cases producing the same +2% return

$$\text{today's return} = \text{today's volatility} \times \text{today's std. shock}$$

Hypothetical case	Volatility	Shock
Calm, big shock	1%	+2
Turbulent, normal shock	2%	+1

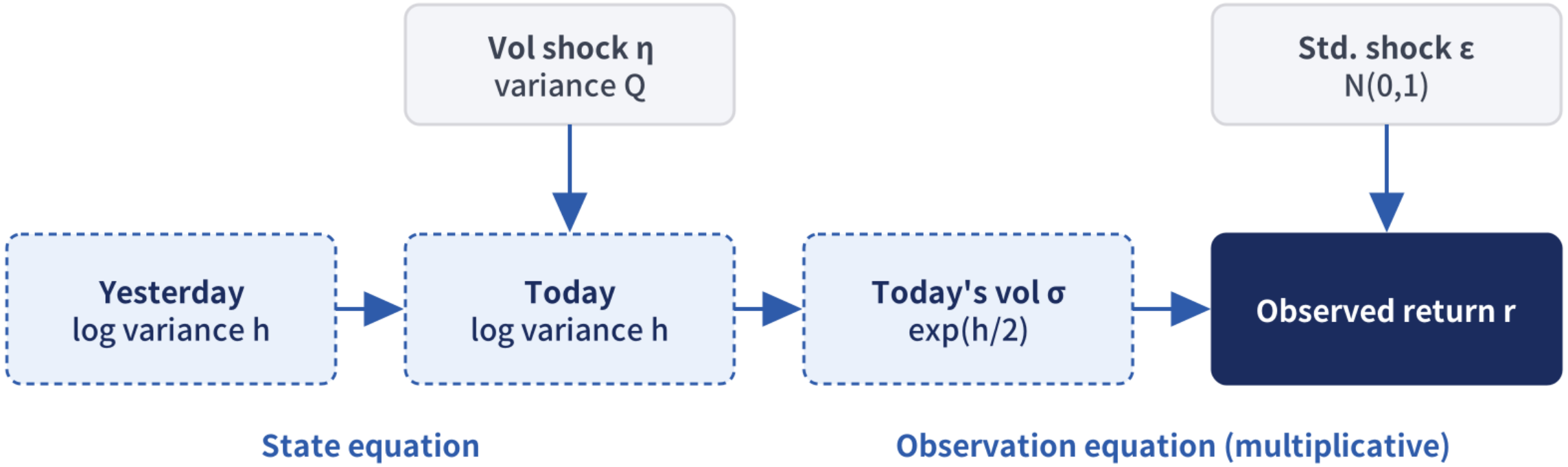


So

- Use many days of moves together with the persistence of the state equation; large drops also signal high volatility

Return equals volatility times a standardized shock

The observation equation is multiplicative, so the state changes the return variance



Squaring and taking logs makes the equation additive

With the original SV observation equation the standard filter cannot compute the exact posterior

Square and take logs

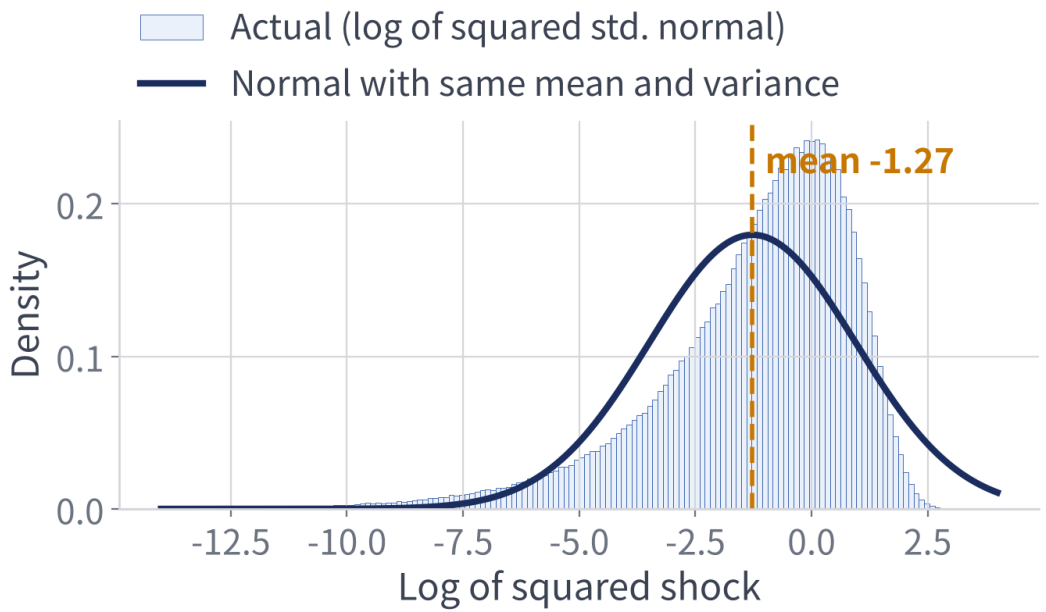
$$\log(r_t^2) = h_t + \log(\varepsilon_t^2)$$

Constants of a standard normal shock

$$c = E[\log(\varepsilon_t^2)] \approx -1.2704, \qquad \mathrm{Var}[\log(\varepsilon_t^2)] = \frac{\pi^2}{2} \approx 4.9348$$

Definitions

$$z_t = \log(r_t^2) - c, \qquad u_t = \log(\varepsilon_t^2) - c$$



The transformed model is linear but not Gaussian

A normal with the same mean and variance links filter and smoother · QML approach

Transformed state eq.

$$\underbrace{h_t}_{\text{log variance}} = \underbrace{\mu_h}_{\text{long-run mean}} + \underbrace{\phi}_{\text{persistence}}(h_{t-1} - \mu_h) + \underbrace{\eta_t}_{\text{vol shock}}, \qquad \eta_t \sim N(0, Q)$$

Transformed obs. eq.

$$\underbrace{z_t}_{\text{transformed obs.}} = h_t + \underbrace{u_t}_{\text{non-normal error}}, \qquad E[u_t] = 0, \quad \mathrm{Var}(u_t) = \pi^2/2$$

Interpretation

- The filter does not estimate the original SV model exactly

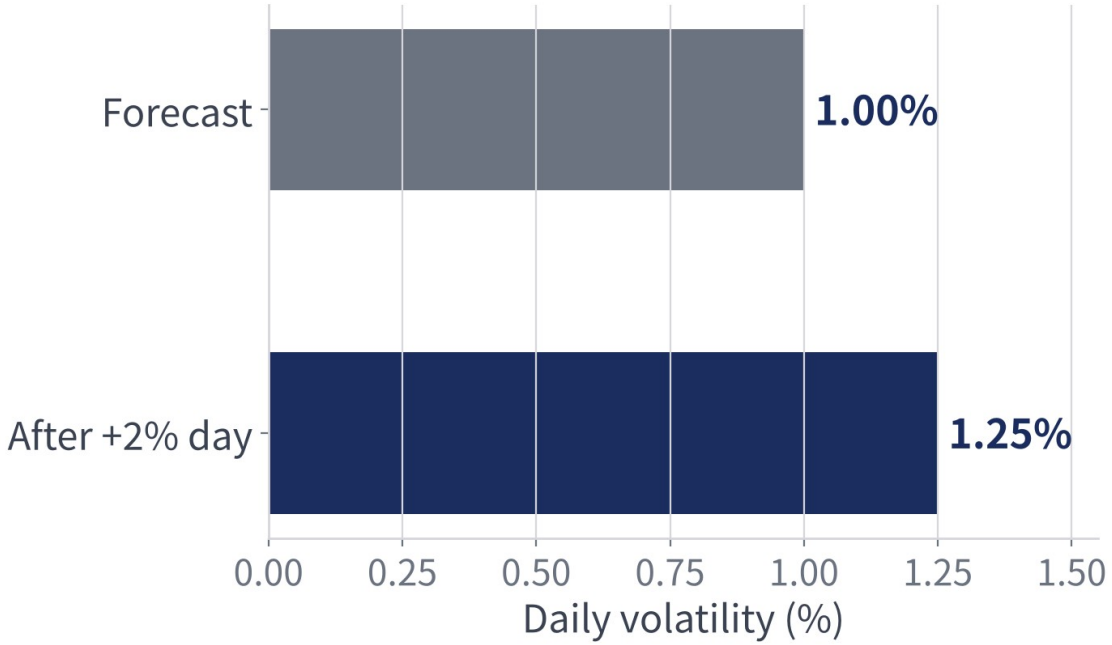
Implementation

- Zero returns cannot be logged; record the adjustment and the return unit

A +2% day lifts the volatility estimate to 1.25%

Hypothetical · forecast error variance 1 · a −2% day gives the same update · see workbook

$$\hat{h}_t^- \approx -9.2103 \;\rightarrow\; \hat{h}_t \approx -8.7627, \qquad \hat{\sigma}_t = \exp(\hat{h}_t/2) \approx 1.25\%$$



Under the normal-approximation posterior

$$\underbrace{\exp(\hat{h}_t/2)}_{\text{median volatility}} \qquad \underbrace{\exp(\hat{h}_t/2 + P_t/8)}_{\text{posterior mean volatility}}$$

Nonlinear transform trap

- Transforming the mean differs from averaging the transformed values

In the normal-approximation model, learn Q first

This EM maximizes the likelihood of the approximate model · the observation error variance is fixed

Step	Output
E-step	Mean, uncertainty and neighbor link of log variance
M-step	New Q from expected squared state residuals

$$\underbrace{Q_{\text{new}}}_{\text{log-var. shock var.}} = \frac{1}{T-1}\sum_{t=2}^{T} E\left[\{h_t - \mu_h - \phi(h_{t-1} - \mu_h)\}^2 \mid z_1, \ldots, z_T\right]$$

Extension · also update persistence

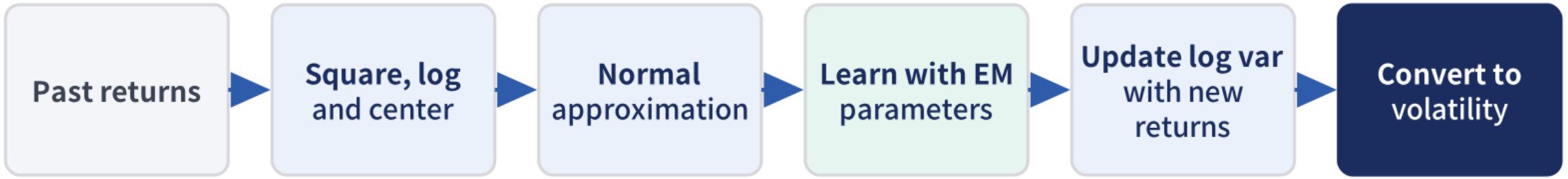
$$h_t = a + \phi h_{t-1} + \eta_t, \qquad \mu_h = a/(1-\phi)$$

More precise methods

- Mixture of normals, particle filter or simulated EM, Bayesian MCMC

SV estimation starts and ends with a transform

Hedging tracks co-movement, SV the size of moves · combining both is harder



Lab · Estimate volatility from SPY returns

Big picture: log variance is the state, the transformed return observed, normal approximation

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```
Run the stochastic volatility lab. Work folder: ssm_lab/sv.

1. Download SPY adjusted prices since 2010 with yfinance,
   compute daily log returns and remove the mean.
2. Transform z(t) = log(r(t)^2) + 1.2704 and decide and
   record how to handle days with zero return.
3. State h(t) = mu + phi*(h(t-1) - mu) + eta
   Observation z(t) = h(t) + u, Var(u) = pi^2/2 (normal approx.)
4. Estimate h with the Kalman filter and smoother and learn Q
   with EM. Also build an extended version that estimates mu, phi.
5. Charts: |return| vs estimated volatility exp(h/2);
   comparison with 21-day realized volatility.
   Summarize in report.md and open it. Fix errors yourself.
```

Output and checks

- Output: transformed data, volatility series, Q (and mu, phi in the extension), two charts
- Does estimated volatility spike in March 2020?
- Does the report state this is a normal-approximation estimate?

06

PART 6 · Heston volatility

Option-model variance as a state

Mean-reverting variance, correlated shocks and the need for nonlinear filters

The Heston model uses variance itself as the state

Variance is volatility squared · Δt is the time step

State equation

$$\underbrace{v_t}_{\text{today's variance}} = v_{t-1} + \underbrace{\kappa(\theta - v_{t-1})\Delta t}_{\text{reversion to long-run level}} + \underbrace{\sigma_v \sqrt{v_{t-1}}\,\eta_t}_{\text{variance shock}}, \qquad \eta_t \sim N(0, \Delta t)$$

Observation equation

$$\underbrace{r_t}_{\text{return}} = \left(\mu - \frac{1}{2} v_t\right)\Delta t + \underbrace{\sqrt{v_t}\,\varepsilon_t}_{\text{return shock}}, \qquad \varepsilon_t \sim N(0, \Delta t)$$

Difference from SV

- SV uses log variance; Heston uses variance as the state

Key feature

- Return shocks and variance shocks need not be independent

Heston combines reversion and correlated shocks

Volatility rising when prices fall sharply is called the leverage effect

Symbol	Meaning
v_t	Today's hidden variance · volatility squared
θ	Long-run variance level
κ	Speed of reversion to the long-run level
σ_v	How much variance itself fluctuates (vol of vol)
ε_t	Return shock
η_t	Variance shock
ρ	Correlation between return and variance shocks

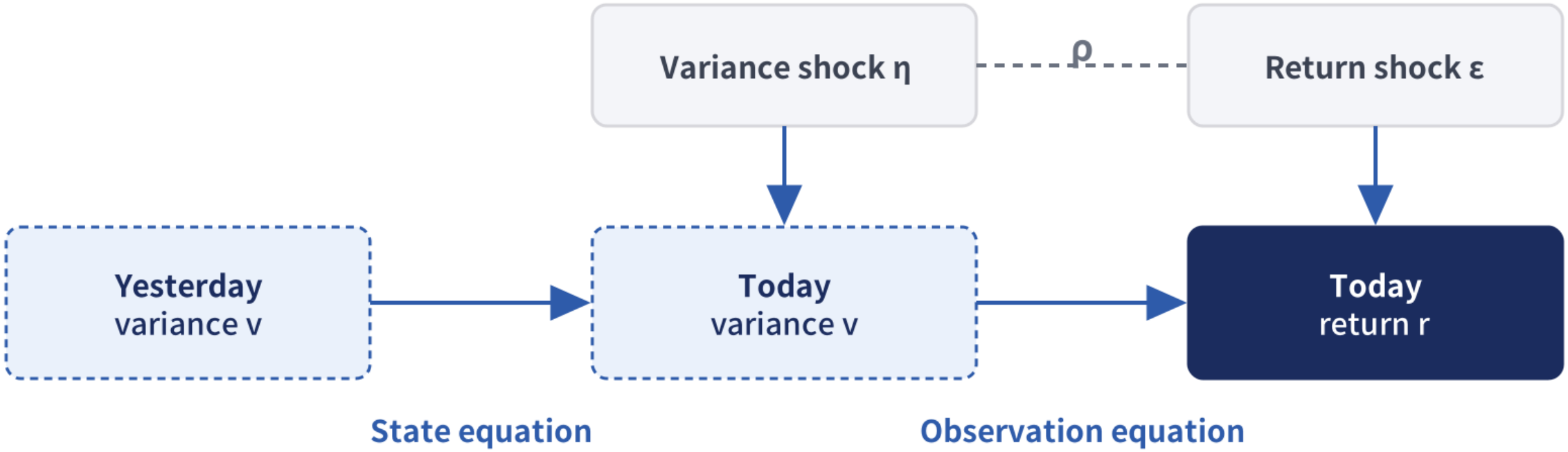
$$\mathrm{Corr}(\underbrace{\varepsilon_t}_{\text{return shock}}, \underbrace{\eta_t}_{\text{variance shock}}) = \rho$$

Leverage effect

- Volatility rises when prices fall sharply
- The correlation is then usually read as negative

Variance and return shocks are linked by correlation

The dashed line shows the correlation between the two shocks



A standard Kalman filter cannot solve Heston

The spending, rate and hedging examples had linear observation equations with normal errors

Difficulty	Reason
Nonlinear	Returns use the root of variance
State-dependent	Shock size scales with the state
Positivity	Variance cannot be negative
Correlation	The two shocks are correlated

Method	Plain explanation
EKF	Linearizes at the current estimate
UKF	Pushes a few sigma points through
Particle filter	Tracks many possible variance paths
Option-based	Adds option prices and implied vol

A straight-ruler filter meets a curved, positive-only problem

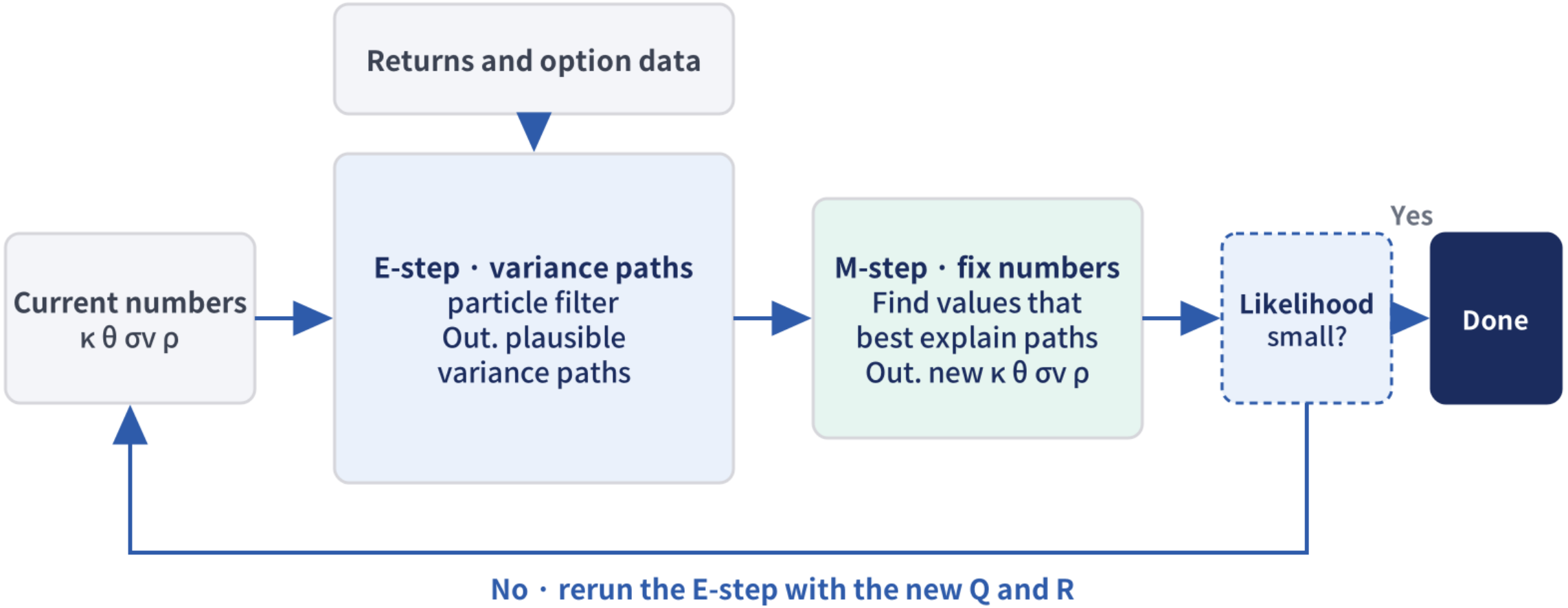
Heston's core structure is still a state space model

EKF = extended Kalman filter, UKF = unscented Kalman filter

Basic state space model	Heston example
State	Hidden variance
Observation	Returns, or option prices and implied volatility
State equation	Variance reverts to its long-run mean
Observation equation	Variance explains returns or option prices
Kalman-type filters	Estimate today's variance sequentially
EM-type methods	Iterate on reversion speed, long-run level, vol of vol and correlation

Heston's E-step simulates variance paths

No closed form · with options the observation equation becomes a Heston pricer



Learn by raising the nonlinearity step by step

Suggested learning order

1. Basic SV

Estimate log variance with a normal approximation

2. Heston

Learn that variance reverts to a long-run level

3. Add option prices

See how much more nonlinear the estimation becomes

Lab · Track Heston variance with a particle filter

Big picture: variance is the state, return observed, a particle filter for nonlinearity · simulated

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```
Run the Heston volatility lab. Work folder: ssm_lab/heston.

1. Simulate a 5-year daily path with kappa=3, theta=0.04,
   sigma_v=0.5, rho=-0.7, mu=0.05.
   Decide how to keep the variance from going negative.
2. Implement a particle filter (2,000 particles) that
   estimates the hidden variance from returns only.
3. Plot the true variance path, the particle filter estimate
   and its 90% band.
4. Apply the SV normal-approximation Kalman filter to the same
   data and compare the estimation errors of both methods.
5. Summarize in report.md and open it with the charts.
   If anything fails, fix it yourself and run to the end.
```

Output and checks

- Output: simulated path, particle filter estimate with 90% band, method comparison table
- Does the true variance path stay mostly inside the 90% band?
- Next, ask it to rerun with rho = 0 and compare the results

07

PART 7 · Nowcasting

Estimating the current cycle

Estimate today's economy from monthly and weekly data before official GDP is released

Nowcasting estimates the cycle from many indicators

Estimating the current state of the economy before official GDP is released

State equation

$$\underbrace{g_t}_{\text{cycle state}} = A\, g_{t-1} + \underbrace{w_t}_{\text{cycle shock}}, \qquad w_t \sim N(0, Q)$$

Observation equation

$$\underbrace{y_t}_{\text{indicators}} = H_t\, g_t + \underbrace{v_t}_{\text{indicator noise}}, \qquad v_t \sim N(0, R)$$

State

- Today's hidden business-cycle state or GDP growth trend

Observations

- Production, retail sales, exports, jobs, prices, finance

Many observations make the equation a vector

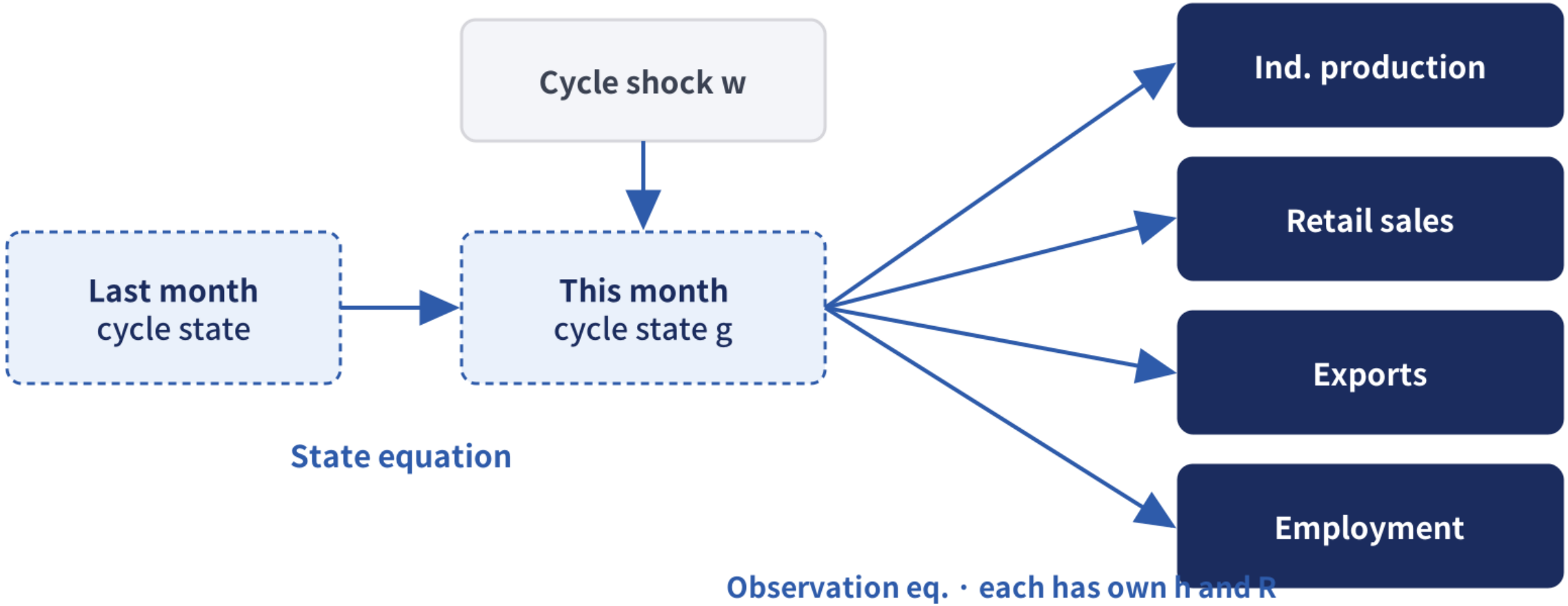
The spending example had a single observation

$$\begin{bmatrix} \text{Ind. production}_t \\ \text{Retail sales}_t \\ \text{Exports}_t \\ \text{Employment}_t \end{bmatrix} = \begin{bmatrix} h_1 \\ h_2 \\ h_3 \\ h_4 \end{bmatrix} g_t + \begin{bmatrix} v_{1t} \\ v_{2t} \\ v_{3t} \\ v_{4t} \end{bmatrix}$$

Symbol	Meaning
g_t	Today's hidden cycle state
y_t	Several economic indicators
H_t	How much each indicator reflects the cycle
w_t	New change in the cycle state
v_t	Indicator-specific noise or measurement error

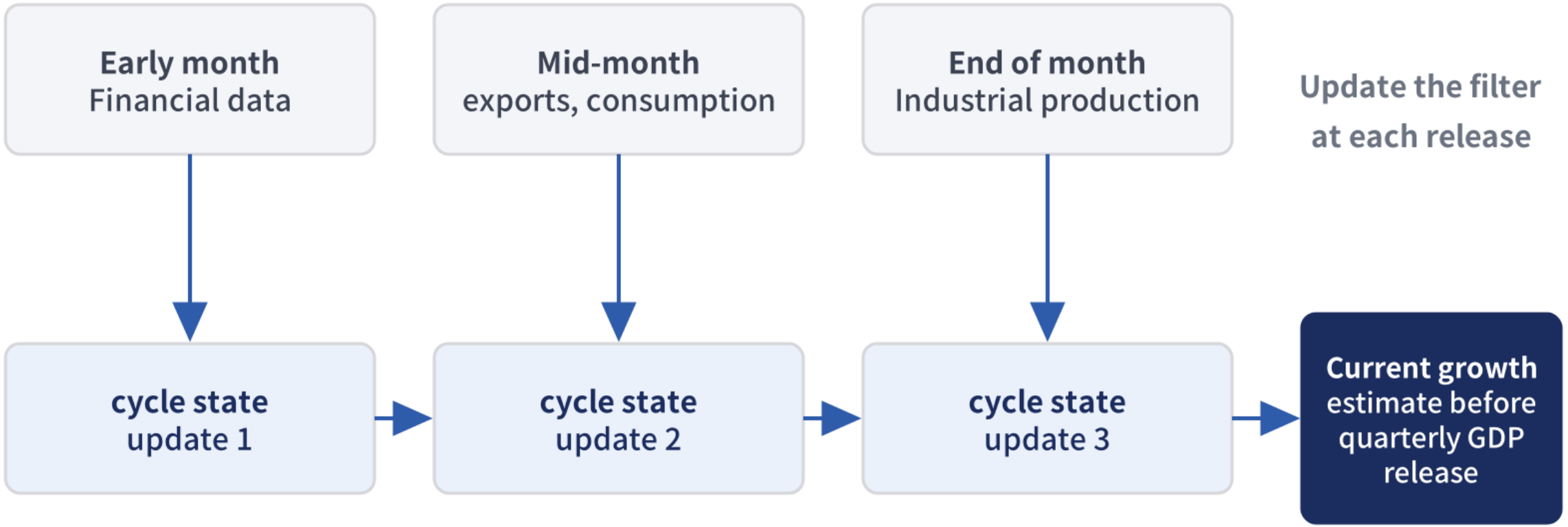
One common cycle state shows up in many indicators

Indicators differ in how much they reflect the cycle and in their noise



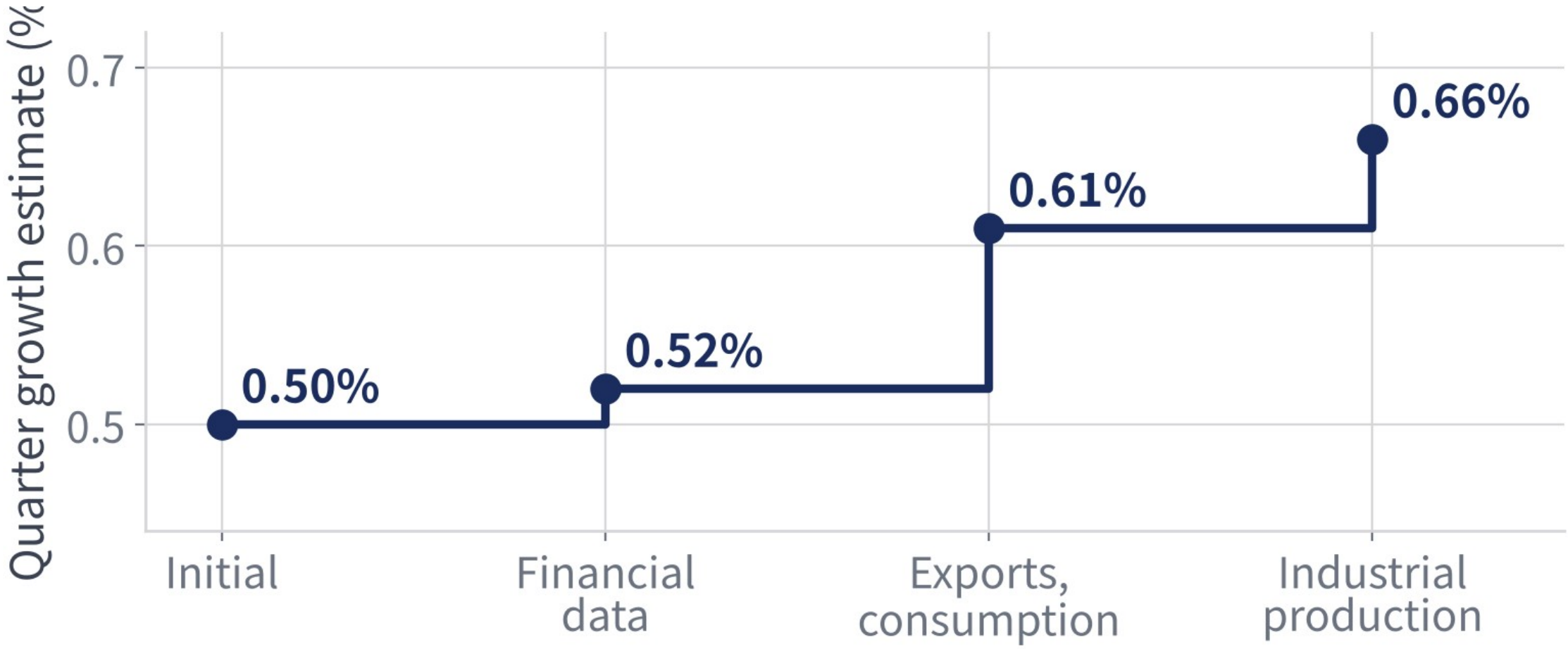
Each release redraws the estimate of the cycle

Indicators are not released at once · some come early, some late



Strong indicators lift the growth estimate

Quarter growth first seen at 0.5% · weak jobs or consumption would lower it · ※ illustrative



Missing and mixed-frequency data fit naturally

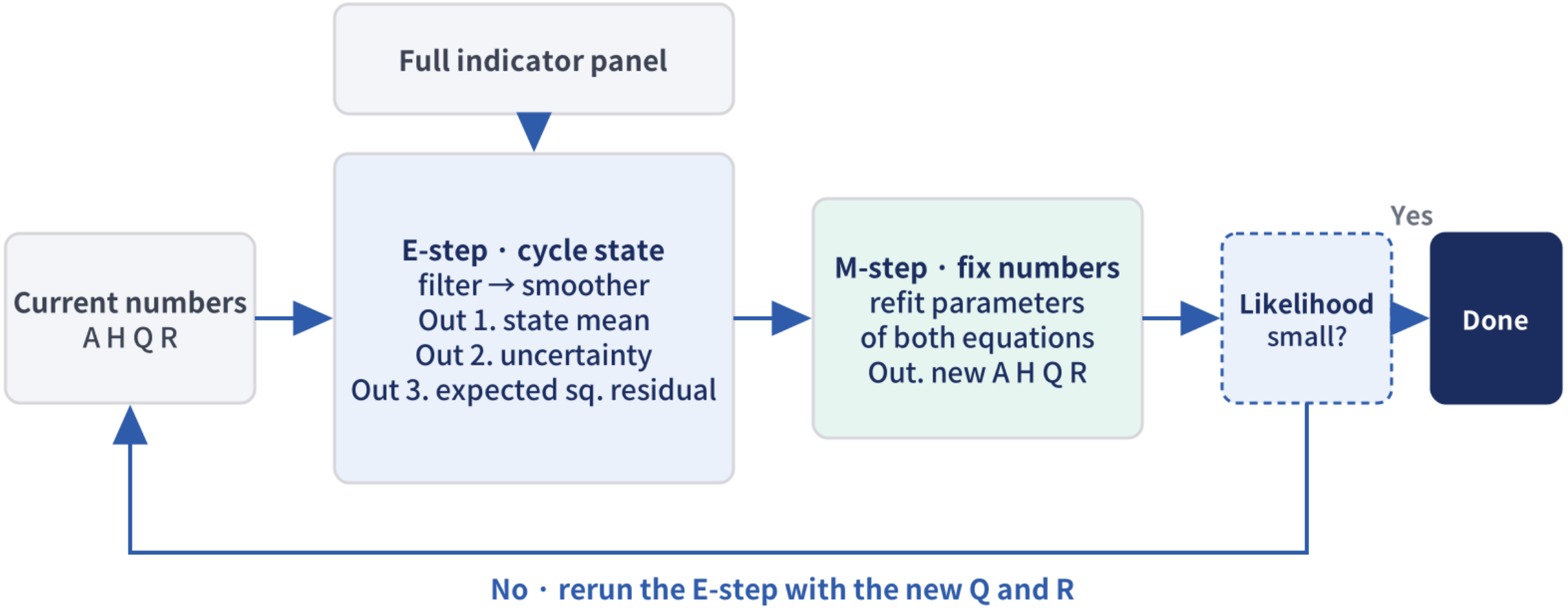
Handling gaps in the data is a key strength of nowcasting

Problem	State space treatment
This month's employment not yet out	Skip that row's observation update
GDP quarterly, indicators monthly	Stack monthly states and link them to quarterly GDP
Different release dates	Update the filter at each release
Different noise by indicator	Reflect in the diagonal or covariance of R

The filter redraws the economic picture as each piece of news arrives

Nowcasting EM updates four numbers

It learns persistence A, loadings H, cycle variance Q and indicator noise R



Joint moves mean the cycle; lone spikes mean noise

The four parameters EM learns and their intuition

Parameter	Meaning
A	Persistence of the cycle state
H	How much each indicator reflects the cycle
Q	Volatility of the cycle state itself
R	Noise or idiosyncratic variance of each indicator

Intuition

- When many indicators move together, the common cycle is estimated more strongly
- When one indicator spikes alone, more of it is read as its own noise
- Indicators that track the cycle get large H
- Indicators that often miss get large R

Lab · Nowcast US GDP from monthly indicators

Big picture: a common cycle is the state, many indicators observed, monthly and quarterly mixed

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```
Run the nowcasting lab. Work folder: ssm_lab/nowcast.

1. Download from FRED since 2000 the monthly INDPRO (industrial
   production), RRSFS (real retail sales), PAYEMS (employment),
   BOPTEXP (exports) and the quarterly GDPC1 (real GDP).
2. Convert indicators to m/m growth and GDP to q/q growth.
3. Estimate a mixed-frequency model with one common factor
   using statsmodels DynamicFactorMQ (EM).
4. Nowcast GDP growth for the latest quarter and show how the
   estimate changes as each indicator is released (news),
   in a table and a chart.
5. Tabulate each indicator's loading (H) in report.md
   and open it. Fix any errors yourself.
```

Output and checks

- Output: indicator panel, estimated factor, nowcast, news table and chart
- Does it still produce an estimate when recent months are missing?
- Does the factor fall in the 2008 and 2020 recessions?

08

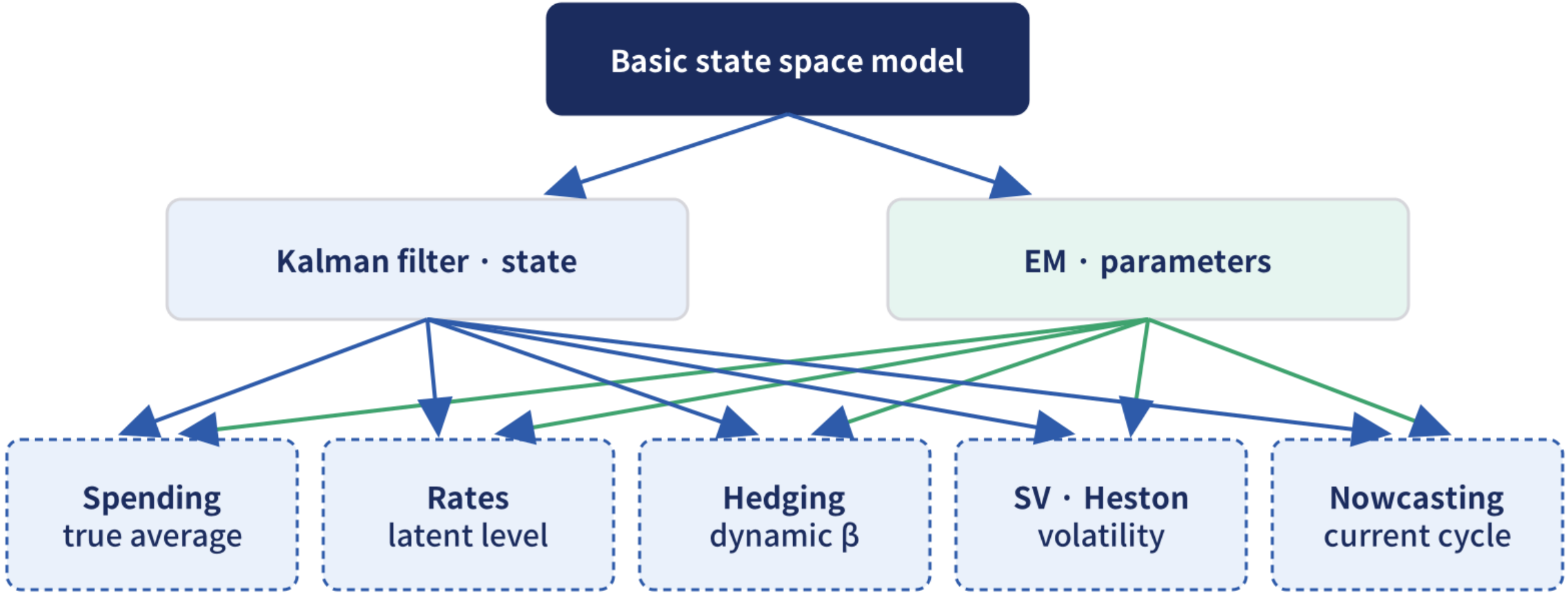
PART 8 · Synthesis

Comparing the examples in one framework

Roles of state, observation, Q, R, Kalman filter and EM

Every example uses the same filter-and-EM framework

Five applications branch out from one basic state space model



The three linear-Gaussian cases differ only in state

All three learn two variances with EM

Item	Spending	Rates	EWC–EWA hedging
State	True average spending	Latent rate level	Hedge ratio
Observation	Survey average	Market rate	EWC return
Link number	1	1	EWA return
What Q captures	Spending change	Level change	Ratio change
What R captures	Survey error	Deviation from level	EWC own move
Kalman filter	Current spending	Current level	Current ratio
EM	Learns two variances	Learns two variances	Learns two variances

Volatility and nowcasting change the observations

SV: normal approximation · Heston: nonlinear filters · nowcasting: many observations

Example	State	Observation	Filter type	Main EM targets
Stochastic volatility	Log variance	Squared, logged returns	Kalman after normal approx.	Long-run mean, persistence, Q
Heston volatility	Variance	Returns, option prices, implied vol	EKF, UKF, particle filter	Reversion speed, long-run level, vol of vol, correlation
Nowcasting	Cycle state	Many indicators	Update at each release	A, H, Q, R

Two equations and two tools form one framework

The core sentence of this lecture

The state equation says how an unseen quantity moves over time; the observation equation says how it shows up in data

Kalman filter

- Estimates the state and its uncertainty with each observation

EM

- Repeats: estimate states in the E-step, fix Q and R in the M-step

A

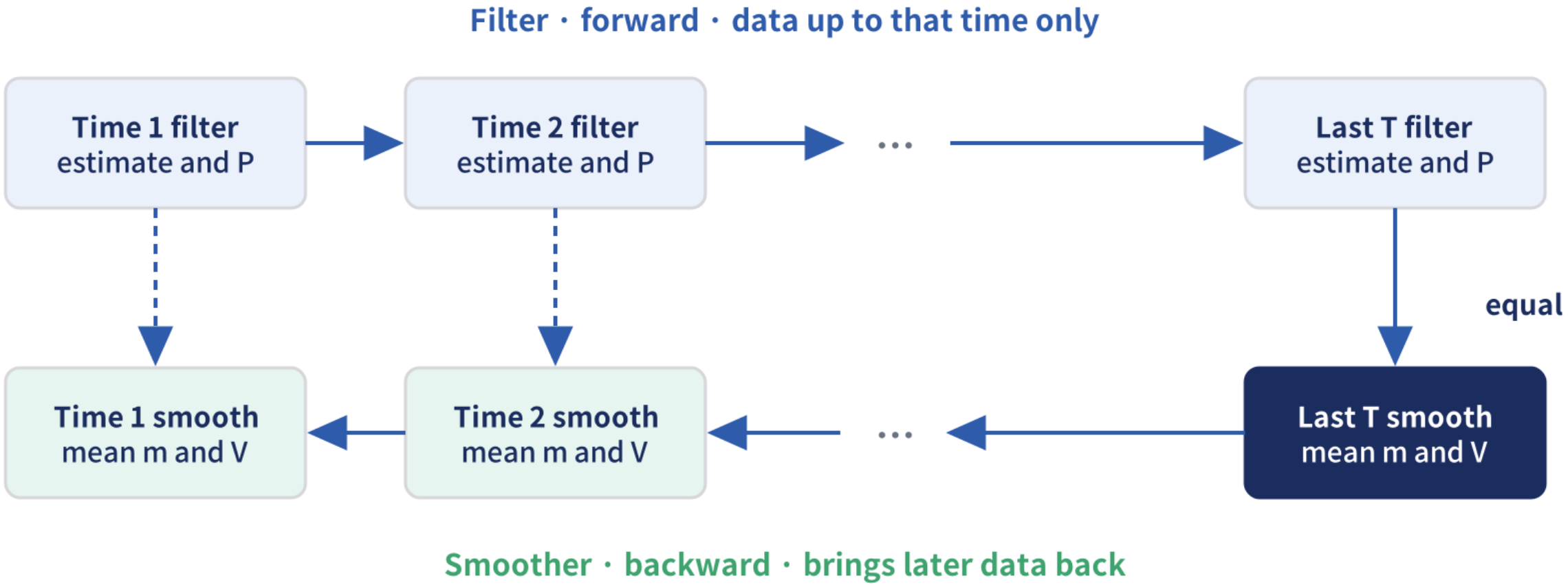
Appendix · Kalman smoother

How is the smoothed variance V computed?

Inside the E-step, filter results are recomputed from back to front

The smoother uses later data to refine the past

Starting from the last time point, it works backward using the filter's estimates and P



Like revising past judgments with later facts

Business-cycle dating is also more accurate when revisited with a few more months of data

The filter's judgment

Made with data up to that time only; uncertainty P remains

What we learn later

If the next estimate differs from its forecast, the earlier judgment is revised in that direction

So V is smaller than P

More data means less uncertainty; at the last time there is nothing more to see, so V equals P

The smoother repeats four lines from back to front

For A = 1 · standard RTS (Rauch–Tung–Striebel) smoother · beyond the original notes

$$m_T = \hat{x}_T, \quad V_T = P_T \qquad \text{starting point}$$

$$J_t = \frac{P_t}{P_{t+1}^-} \qquad \text{pull-back ratio}$$

$$m_t = \hat{x}_t + J_t\,(m_{t+1} - \hat{x}_{t+1}^-) \qquad \text{smoothed mean}$$

$$V_t = P_t + J_t^2\,(V_{t+1} - P_{t+1}^-) \qquad \text{smoothed variance}$$

How to read

- The last time starts from the filter
- Needs only stored estimates, P and forecasts
- The bracket is the next step's drop in uncertainty
- That drop passes back scaled by J squared

The pull-back ratio J compares two uncertainties

The larger J, the more next-step information is pulled back to revise the earlier estimate

$$J_t = \frac{\overbrace{P_t}^{\text{today's filter uncertainty}}}{\underbrace{P_{t+1}^{-}}_{\text{tomorrow's forecast uncertainty}}}$$

Large J

- Today's filter is still uncertain
- Lean on the next step, revise a lot

Small J

- Today's filter is already confident
- Barely revise even with new information

Spending example: last month's variance falls to 6

Last month 300 · 8, this month's forecast 300 · 16, after update 306 · 8 · this month is last

1. Start $m_T = 306, \qquad V_T = P_T = 8$

2. Ratio $J = \frac{8}{16} = 0.5$

3. Mean $m_{T-1} = 300 + 0.5 \times (306 - 300) = 303$

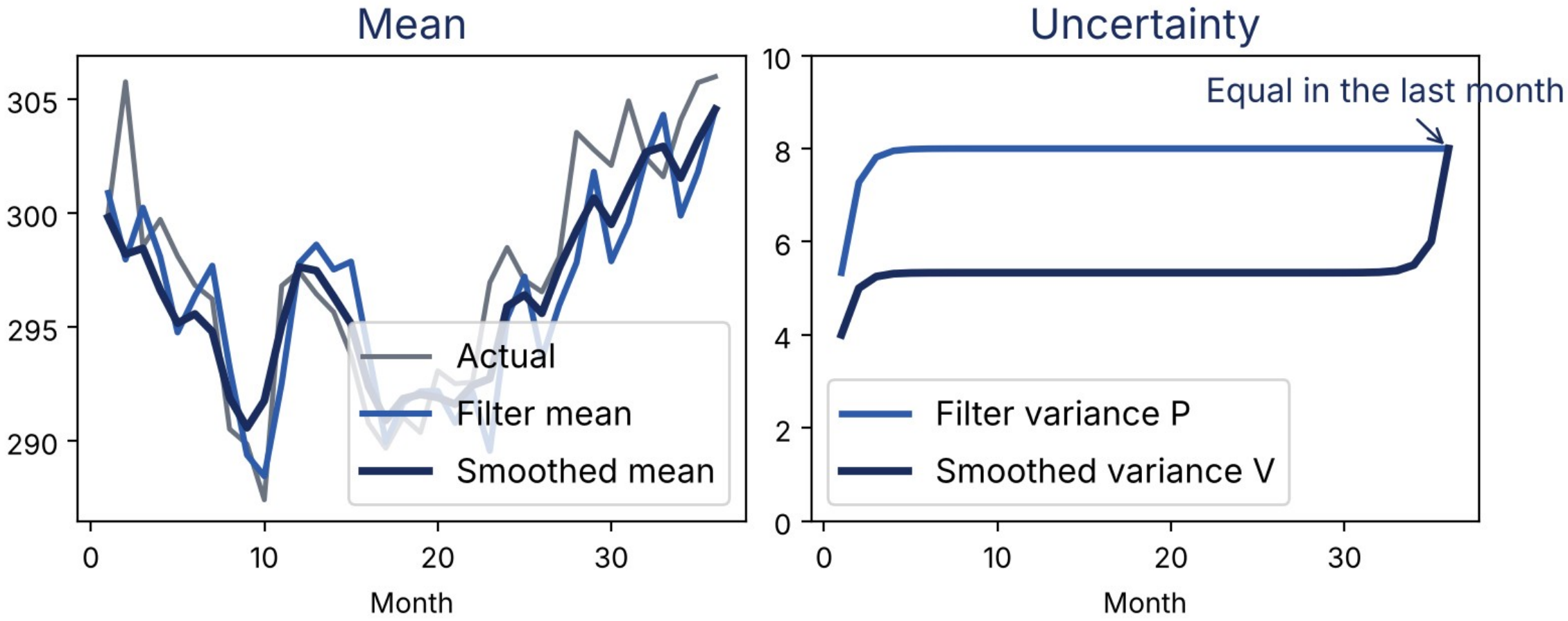
4. Variance $V_{T-1} = 8 + 0.5^2 \times (8 - 16) = 6$

How to read

- Seeing this month rise to 306, last month's estimate is raised to 303
- Having seen one more month, last month's uncertainty falls from 8 to 6

Smoothed variance stays below filter variance

36 simulated months, Q = 8, R = 16 · equal in the last month · ※ illustrative data



Same formulas for hedging; V changes every round

V is an E-step output and the input for the expected squared residuals used in the M-step

What changes in hedging

- The filter's link number is the day's EWA return
- The backward formulas do not contain the link number
- On days with EWA return near 0 the filter variance barely falls, so V stays large

Place inside EM

- V is an E-step output computed with the current Q and R
- When the M-step fixes Q and R, V is recomputed in the next round
- This V is the uncertainty added to the expected squared residuals

References

ETF product information and stochastic volatility estimation literature

Source	Content
BlackRock · iShares MSCI Canada ETF (EWC)	Official EWC product page
BlackRock · iShares MSCI Australia ETF (EWA)	Official EWA product page
Chad Fulton · Stochastic volatility, quasi-maximum likelihood	QML estimation of SV
Kim, Shephard and Chib (1998)	Stochastic Volatility, Likelihood Inference and Comparison with ARCH Models